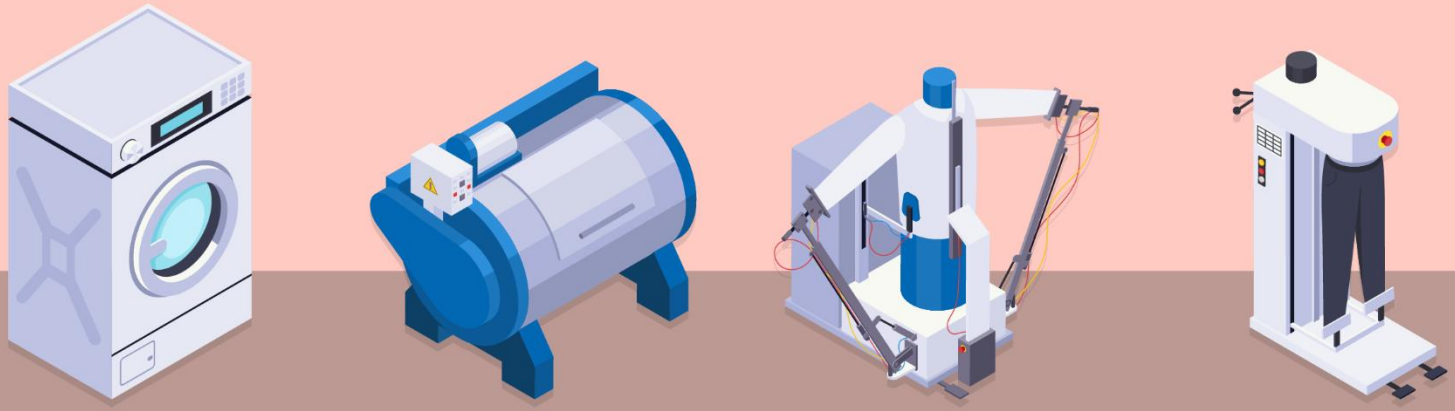


APPAREL WASHING, DYEING & PRINTING

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Apparel Washing

Apparel washing is a technology that is applied to remove dirt, dust; change or modify the outlook, appearance, comfortability, and design of garments through chemical reaction, mechanical friction, or rubbing.

Normally washing means cleaning something. But in the case of garments, washing is not limited to this function. For garments, washing is not only required for cleaning but also required for improving fabric quality, increase the softness and color fastness of fabric, reduce shrinkage, removing dirt, and for creating color effects. It is normally done after stitching. Wash types usually depend on the product nature and its usage, consumer demand, and fashion trends.

Objectives

- To remove dirt, dust, and waste materials from garments.
- To remove size materials from garments.
- To make directly wearable garments after purchase.
- To increase the softness, smoothness of garments.
- To increase color fastness property.
- To increase the brightness of garments.
- To change the appearance of garments.
- To produce color fading effect.
- To create new color and tinted effect on garments.
- To remove hairiness of garments to make it comfortable to wear.
- To meet buyers requirement.

Classification

1. Wet Process/Chemical Process

- Normal wash
- Pigment wash
- Caustic wash
- Bleach wash
- Enzyme wash
- Stone wash
- Stone enzyme wash
- Acid wash
- Super white wash

2. Dry Process/Mechanical Process

- PP spray
- PP sponging
- Sand blasting
- Hand sand
- Wrinkle
- Whiskering
- Tagging
- Grinding
- Destroy

Advantages

- Removal of starch, making the garments soft feeling.
- Dirt and spots present in the garments can be removed by applying garment washing.
- No possibility of further shrinkage after applying garment washing.
- Washed garments can be used directly after purchase from stores.
- Different outlooks can be produced by applying garments washing.
- Similar outlook can be produced by applying different washing techniques.
- Different types of washing produce different effect of outlook.
- Comparatively lower capital investment to set up a garment washing plant.
- Lower land space and least manpower cost are required to run a washing plant.

Fading Effect of Washing

Color fading effect in garment washing refers to the process of intentionally fading the color of a garment to achieve a particular look or style. This effect is typically achieved through various washing and finishing techniques that remove some of the dye or pigment from the fabric.

Color fading can be used to create a range of different effects, from a subtle, worn-in look to a more dramatic, distressed appearance. It can be achieved through a variety of washing techniques, including stone washing, enzyme washing, bleaching, and acid washing etc. Each of these techniques has its own unique characteristics and can be used combinedly to create different types of color fading effects.

Unwashed or Raw Denim

Raw denim, also known as unwashed denim, is a type of denim fabric that has not undergone any washing processes. While manufacturing jeans, they undergo sizing process which involves adding starch to the warp yarn to give it strength and crisp finish. Sizing causes the fabric to become stiff and rough, making them uncomfortable to wear until they have been washed. Washing the denim removes size material and softens the fabric, making it more comfortable to wear. Unwashed jeans can be unhygienic. They can harbor bacteria, sweat, and dirt, which can lead to skin irritation and even infections. Raw denim can also bleed dye onto other fabrics, particularly light-colored fabrics, until it has been washed a few times.

Home & Factory Wash

Home wash refers to washing clothes in a domestic setting, using detergents and household washing machines or by hand. It is usually done on a small scale, with individual garments or small loads of laundry being washed at a time. It is often less rigorous and may not be as effective as factory wash methods, but it is more convenient and accessible.

Factory wash, on the other hand, refers to the washing and creating fading effect of garments in a commercial or industrial setting, using specialized equipment and chemicals. It is usually done on a large scale, with hundreds or even thousands of garments being washed at a time. It is typically more rigorous and efficient than home wash methods, and it is designed to ensure that garments are cleaned thoroughly and consistently.

Denim

Denim & Woven Fabric

Denim is a special type of woven fabric that has its own unique features. Here are the main differences that make denim stand out:

Twill Weave: Unlike other woven fabrics, denim is made using a twill weave, which creates a diagonal pattern on the fabric surface. This makes denim stronger and more durable.

Indigo Dyeing: Denim is traditionally dyed with indigo, giving it a classic blue color. Over time, the color naturally fades, creating the worn-in look we associate with denim.

Weight and Thickness: Denim is heavier and thicker compared to other woven fabrics. This makes it perfect for making sturdy clothing like jeans and jackets.

Texture: The twill weave and indigo dye give denim a unique texture and appearance. It has a slightly rough surface and a faded appearance.

In summary, denim stands out because of its twill weave, indigo dyeing process, weight, texture, and its common use in clothing items. It is primarily used for making jeans, jackets etc.

Popularity of Indigo Dyed Denim

Indigo dye is what makes denim so captivating. It gives the fabric a beautiful blue color that has been cherished for centuries. One special thing about indigo dye is that it gradually fades over time, creating a unique and personalized look with wear and washing. This natural aging effect is loved by denim enthusiasts who appreciate the worn-in appearance. With indigo dye, we can achieve different shades of blue, from deep and dark to lighter and more faded.

Indigo dye is very durable and can withstand repeated washings without fading. Indigo dye is relatively inexpensive, which makes it a cost-effective option for denim manufacturers. Unlike many other dyes, indigo dyed denim does not pose health hazards. Overall, the popularity of indigo dye in denim can be attributed to its unique fading properties, versatility, durability, cultural significance, and timeless appeal.

Characteristics of Indigo Dye

- Indigo is a dark blue color.
- Indigo is insoluble in water. It must be reduced to a soluble form before it can be used.
- Indigo is a very fast dye that gives fabrics a long-lasting color.
- Indigo is relatively inexpensive.
- Indigo is easy to use, but it can be difficult to apply evenly.
- Indigo can sometimes bleed when washed.
- Indigo dye is a versatile dye that can be used to dye a variety of fabrics.
- Indigo dye is a natural dye that is environmentally friendly and sustainable.

Washing Parameters

There are four important factors to consider when washing apparel: chemical concentration, time, temperature, and M:L ratio. Each of these factors plays a role in achieving effective and safe washing results.

Chemical Concentration

Chemical concentration refers to the amount of detergent or other cleaning agents used in the washing process. The concentration of the cleaning agent is determined by the degree of washing effect on the garment and the type of fabric. Using too much or too little detergent can affect the cleaning efficiency and can cause damage to the fabric.

Temperature

The water temperature used in the wash impacts the cleaning effectiveness and desired wash effects. It's best to keep the temperature between room temperature and 60ºC. Higher temperatures can weaken garments and cause higher color fading.

Time

The duration of the washing process is also crucial. Heavy garments may require a longer washing time, while delicate fabrics need shorter durations. Generally, washing times can range from 10 to 60 minutes.

M:L Ratio

This ratio refers to the weight of the garments compared to the weight of the water in the washing machine. It ensures thorough washing and rinsing. The ideal M:L ratio is between 1:8 and 1:10, although ratios up to 1:20 can be used.

Process Flow

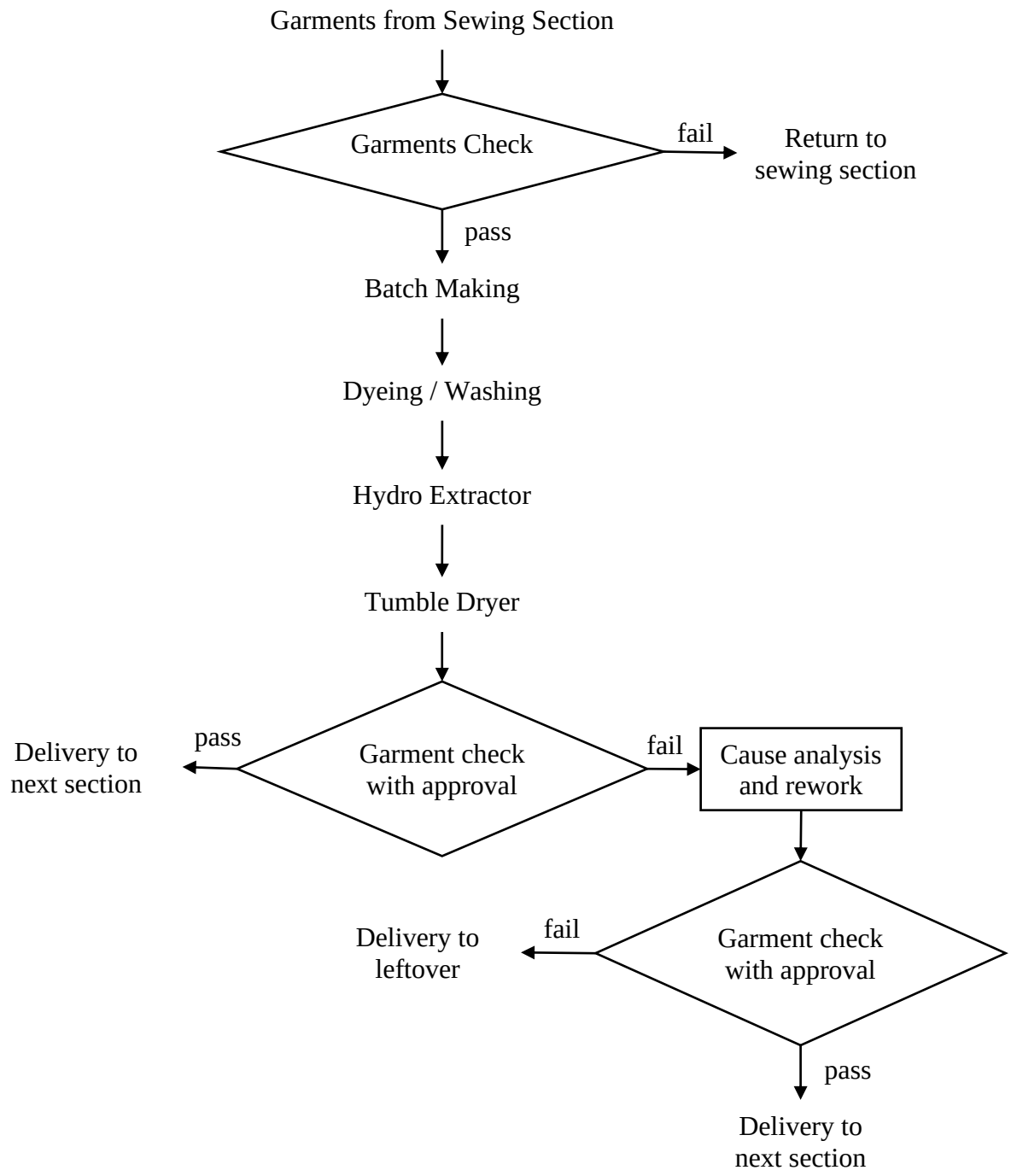
Batch Preparation
↓
Desizing — Proper penetration of chemicals in further process. 10-15 minutes. 50ºC-60ºC.
↓
Hot Wash — Stop redeposition of sizing materials. 5 minutes. 70ºC-80ºC.
↓
Cold Wash — Bring down to washing room temperature. 3-5 minutes. 40ºC-41ºC.
↓
Designated Wash
↓
Hot Wash
↓
Cold Wash
↓
Neutralization — For bleach wash. Not for enzyme wash.
↓
Hot Wash
↓
Cold Wash
↓
Softening — No further washing after this process.
↓
Hydro Extracting
↓
Drying
↓
Quality Checking
↓
Finishing

Washing Procedure

Batch Preparation
↓
Loading Garments
↓
M:L Ratio
↓
Adding Chemicals
↓
Setting Temperature
↓
Setting Time
↓

Dropping Liquor
↓
Washing
↓
Softening
↓
Unloading
↓
Hydro Extracting
↓
Drying

Factory SOP



Washing Machineries

Washing Machine: Used for the main washing process, these machines agitate the garments with water, detergent, and other chemicals to remove dirt, oil, and stains.

Hydro Extractor: This machine is used to remove excess water from washed garments through centrifugal force, reducing drying time and energy consumption.

Dryer: Used to dry the washed garments after the washing process, these machines employ heated air circulation to remove moisture.

Boiler: Provides steam for different types of wash like enzyme, bleach; various processes like and fixing, softening, wrinkle removing or applying heat during drying.

Sample Washing Machine: These machines are used for small-scale washing and testing purposes, allowing for the evaluation of different processes and effects before large-scale production.

Effluent Treatment Plant (E.T.P): Clean waste water and remove any toxic and non-toxic materials or chemicals from it so that that water can be reused or released in the environment which will do less harm to the environment.

Domestic & Industrial Washing Machine

The main difference between domestic and industrial washing machines is their size and capabilities. Domestic machines are designed for personal use and are easier to operate, while industrial machines are built for larger loads and have more advanced features.

Industrial washing machines prioritize stability, performance, durability, and efficiency. They can handle extra-large wash loads ranging from 15kg to 300kg per batch. In contrast, domestic machines have smaller capacities, typically washing between 6 to 12kg per batch, and focus on minimizing noise during operation. Industrial machines are powerful and can effectively clean heavy and bulky fabrics, which domestic machines may struggle with. They also offer features like automated detergent and softener release, as well as programmability options. These advanced features are not commonly found in standard domestic washing machines.

Working Principle of Hydro Extractor

A hydro extractor is a machine used in textile and garment industries for removing excess water from washed fabrics. The working principle of a hydro extractor is based on centrifugal force.

When the machine is turned on, the drum inside the machine starts spinning at a very high speed, typically between 800 and 1600 RPM. The wet garments or fabrics are loaded into the drum, and as it spins, the centrifugal force pushes the water out of the garments or fabrics. The force generated by the spinning drum is so strong that it causes the water to be squeezed out of the fibers of the garments or fabrics. The water is then drained out of the machine through a drainpipe.

Steam Dryer & Gas Dryer

The main difference between a steam dryer and a gas dryer is the method they use to dry clothes. A steam dryer uses hot water vapor to dry clothes. Steam is converted to heat by using a heat exchanger and the heat dries cloth. A gas dryer, on the other hand, uses natural gas or propane to generate heat that dries clothes. The gas is burned in a burner assembly, which heats the air that is circulated through the dryer drum.

Steam dryers are commonly used to finish garments and fabrics. Steam is used to remove wrinkles and creases, soften and relax the fabric. Gas dryers are commonly used for industrial drying. It produces yellowish color effect in white fabric. Gas dryers are often used after washing to remove excess water and prepare the garments for further processing, such as ironing or pressing.

Washing Chemicals

Detergent: Detergent is used to remove impurities and size materials from garments. It is the mostly used chemicals in garment washing plant. Detergent is a water-soluble cleaning agent. It is made from long molecules of sodium or potassium salts that contain a head and a tail. The head of the molecule is attracted to water (hydrophilic) and the tail is attracted to oil, grease and dirt (hydrophobic). It decreases the surface tension (surfactant) of water. pH value of detergent is 7-10. There are three types of detergents: anionic, cationic, non-ionic. Anionic and cationic detergents are mostly used.

Bleaching Powder: Bleaching powder is an oxidizing agent. It is used in washing plant for color fading from the denim garments. Different shade of color can be achieved on garment i.e. dark, medium, light shade.

Caustic Soda: Caustic soda is the strong alkali used in garments washing to introduce an alkaline condition. It creates the look in bleach technique without changing the color of garments. Caustic soda has a cleaning power which is more powerful than soda ash. Fading effect/old looking effect come rapidly on garments by the action of caustic soda.

Soda Ash (Na_2CO_3): It has a cleaning power and helps in color fading from garment during washing. Soda ash creates alkaline medium in the washing bath and breaks pigment dye from fabric surface. It also helps to control uniform bleaching in bleach bath.

Sodium Metabisulfite: Sodium metabisulfite is used in the washing plant to neutralized the garment from potassium permanganate or acidic medium.

Sodium Hyposulfite: Sodium hyposulfite is used to neutralize the garments from chlorine bleach.

Acetic Acid: Acetic acid is a weak organic acid which is used to neutralize the garments from alkaline condition and to control the pH value in wash bath.

Flax Softener (Cationic, Non-Ionic): Derived from natural sources, specifically from the flax plant. Flax softener is used to make the garments soft, silky and also provides excellent lubricating properties.

Silicon Softener: Silicon softener is a textile finishing agent consisting mainly of amino modified silicon. When applied on fabrics, it gives durable softness, lubricity, slippery effect, elastic handle, anti-pilling, dimensional stability, tear resistance and allows fabric to be cut and sewn more easily, improving wear and easy care properties.

Anti Staining Agent: Anti staining agent is used to prevent back staining on weft yarn (white yarn) of the denim and increase the brightness of fabrics.

Desizing Agent: Desizing agent is used to remove mainly starches, cmc, waxes, fats, pectin, minerals and unfixed indigo dye from denim, twills, poplin, and canvas fabrics etc.

Sodium Bicarbonate: Sodium bicarbonate is used in washing plant in the bleach bath with bleaching powder for denim light shade. It helps to come out the color easily within short time. As a result, production increase and costing become low.

Potassium Permanganate: Potassium permanganate is a strong oxidizing agent used in acid wash with pumice stone for color fading from the garments. It is also used in PP spray chamber (by nozzle) for color fading (whitish affect) of the garments.

Hydrogen Peroxide: Hydrogen peroxide is one kind of bleaching agent. In alkaline medium, hydrogen peroxide breaks up and gives some per-hydroxyl ion, which discolor the coloring materials and as a result fading effect is developed. Hydrogen peroxide is used in scouring, bleaching bath for white/ready for dyeing gray fabric garments. It is also used to neutralize the garment from alkaline condition.

Stabilizer: Hydrogen peroxide works in good condition at temperature above 90ºC. When temperature raise to 90ºC, it breaks the hydrogen peroxide. Stabilizer is used to protect the breaking of hydrogen peroxide and peroxide works in bath smoothly.

Fixing Agent: Fixing agent is used for unfixed dye to fix on fabrics. When fabric color is properly fixed, then color fastness and rubbing fastness will be increased.

Sodium Chloride (Salt): It helps to exhaust dye into the fiber.

Buffer: Buffer is used in washing plant for pH control of enzyme bath, softener bath, desizing bath.

Catanizer: Catanizer is used in pigment exhaustion method. Pigment colors have no affinity to fabric. When catanizer is used in fabric dyeing process, it increases the affinity between pigment color and fabrics.

Optical Brightener: Optical brighteners are used for improving the brightness of garments in the washing plant. Red brighteners and Blue brighteners are mainly used as optical brighteners for improve the brightness of garments.

Resin: Textile resin is used for the creation of semi-permanent creases in denim and other cellulose fabrics. It is used for cotton and polyester fabric. After washing, fabric retains soft handle.

Washing Faults

Back Staining: Back staining is a phenomenon of the re-deposition of discharged indigo dye from the process bath to fabric. As a result, bluish color distributed all over the back side of the denim fabric or white pocket fabric. There are two major reasons for back staining problem. One is for excess softener application on garments and another reason is for low quality dyeing process on garments. Back staining problem is also created by the application of low quality dyestuff on garments.

Garments Discoloration: If color of garments or any parts of garments changes after washing it's called discoloration. It is one of the common defects. It can be caused by improper dye fixation, inadequate rinsing, or the use of harsh chemicals that affect the original color.

Shade Variation: Shade variation happens when there are noticeable differences in color or hue within the same batch of garments after washing. It can occur due to inconsistent dye penetration or uneven distribution of dyes during the washing process.

Crease Marks: Crease marks are visible lines or wrinkles that remain on the garment after washing and drying. They can occur due to improper folding or stacking of garments, excessive agitation, or inadequate drying techniques.

Washing Mark: Washing marks are spots or discolorations that appear on garments after they have been washed. They are caused by the garments rubbing against each other or against the washing machine drum. Washing marks can be difficult to remove, and they can make garments look worn and untidy.

Bleach Spot: Bleach spots are areas on the garment where the original color has been bleached or lightened due to the use of bleaching agents during washing. This can occur when the bleach comes into direct contact with the fabric.

Stain Formation: Stains can occur during washing due to the presence of substances like oil, grease, or other chemicals that were not properly removed or treated before the washing process.

After Wash Hole: After wash hole refers to the formation of small holes or tears in the fabric of the garment after washing. This can happen due to excessive mechanical action, abrasion with other garments or washing machine parts, or the use of harsh chemicals.

Poor Hand Feel: Poor hand feel refers to the undesirable tactile sensation or texture of the fabric after washing. It can be caused by excessive use of harsh chemicals, inadequate softening, or improper washing techniques.

Poor Sanding: Poor sanding refers to the unsatisfactory or inconsistent sanding effect on the garment's surface during the washing process. It can result in an uneven or undesirable texture on the fabric.

Poor Grinding: Poor grinding refers to the inadequate or unsatisfactory grinding effect on the garment's surface. It can lead to an inconsistent or undesirable worn-out appearance.

Poor Brightness: Poor brightness refers to a lack of luster or shine on the garment after washing. It can be caused by inadequate rinsing, insufficient use of brightening agents, or improper drying techniques. Poor brightness can make the garment appear dull or faded.

Care Label Damage: Care label damage occurs when the label attached to the garment is damaged or becomes illegible during the washing process. It can happen due to excessive agitation, abrasive contact with other garments or machine parts, or the use of harsh chemicals that degrade the label material. Care label damage can make it difficult for consumers to properly care for the garment in the future.

Bottom Hem and Course Edge Destroy: Bottom hem and course edge destruction refers to damage or deterioration of the garment's hemline or edges after washing. This can happen due to excessive mechanical action, abrasive contact with other garments or machine parts, or weak stitching.

Bad Smell: Poor neutralization can result in a lingering bad odor on the garment after washing. It occurs when the acidic or alkaline substances used in the washing process are not properly neutralized, leading to an unpleasant smell.

Lycra Out: Lycra out refers to the problem of elastic fibers, such as Lycra or spandex, becoming loose or dislodged from the fabric during the washing process. This can happen due to excessive mechanical action, high temperatures, or the use of harsh chemicals that degrade the elasticity of the fibers.

Wet Process

Wet washing process of garment involves the use of water and chemical additives at a definite temperature according to a recipe to alter the appearance of the garment. Wet process is commonly used to soften and smoothen the fabric, remove dirt and stains, and add color effect to the garment.

Why M:L ratio should be low during softening?

M:L ratio in garment washing refers to the ratio of the weight of the fabric (material) to the volume of the water (liquor) used in the washing process. During the softening process of garments washing, it is recommended to keep the M:L ratio low, typically around 1:4 to 1:8, depending on the type of softener and the fabric being treated.

The reason for keeping low M:L ratio is to ensure that the softener is evenly distributed throughout the fabric. If the M:L ratio is too high, the softener may not be able to penetrate the fabric properly, resulting in uneven softening and potentially leaving some areas of the garment feeling stiff or uncomfortable. Additionally, a high M:L ratio may dilute the softener too much, reducing its effectiveness and leading to increased costs.

What will happen if M:L ratio is lowered

M:L ratio in garment washing refers to the ratio of the weight of the fabric (material) to the volume of the water (liquor) used in the washing process. Lowering the M:L ratio means reducing the amount of water used in the washing process, which can have several effects on the outcome of the washing process:

Reduced Water Usage: A lower M:L ratio means less water is used, resulting in lower water consumption and decreased wastewater generation.

Increased Abrasion: With less water in the washing process, the fabric is more likely to rub against the machine's mechanical parts, resulting in increased abrasion and possible fabric damage.

Uneven Distribution of Chemicals: A lower M:L ratio means that the chemicals used in the washing process are more concentrated in the smaller amount of water used. This can result in uneven distribution of the chemicals on the fabric, leading to inconsistent results.

Reduced Effectiveness: As the chemicals become more concentrated in a smaller amount of water, their effectiveness can decrease, resulting in less effective washing of the fabric.

Therefore, while lowering the M:L ratio can reduce water usage, it can also have negative effects on the washing process's quality and outcome. It is important to maintain an appropriate M:L ratio for each type of washing process to achieve the desired results.

What will happen if desizing is not done

Desizing is a crucial step in garment washing that involves removing the sizing agents applied to the fabric during the manufacturing process. If desizing is not done before garment washing, the sizing agents may react with the chemicals used in the washing process, leading to unwanted effects on the garment.

For example, if the garment is dyed, the sizing agents may prevent the dye from penetrating the fabric evenly, resulting in uneven coloration or patchy dyeing. Additionally, if the sizing agents are not removed, they may cause the fabric to become stiff or harsh, reducing the garment's overall quality and comfort. In summary, if desizing is not done before garment washing, it can negatively impact the appearance, texture, and quality of the garment.

Normal Wash



Normal wash is a standard garment washing process that involves the use of water, detergent, and mechanical action to clean and soften the fabric. It is a combination of rinse wash and softener wash. It is suitable for most types of garments made from natural or synthetic fibers. It is an effective way to remove dirt, sweat, and odor from clothes and to keep them looking clean and fresh.

Objectives

- Remove dust, dirt, oil spot, impurities from the garments.
- Remove unfixed dye.
- Remove size materials from the garments.
- Remove starch presents on the garment fabrics.
- Create soft feeling to wear the garments after purchasing.
- Help to achieve buyer washing standard.

Process

Step 1: Normal Wash

- Lot size: 70 kg
- Water: M:L = 1:8 - 10 560-700 liter
- Machine running
- Detergent: 0.5 gm/L 280-350 gm
- Temperature: 40ºC - 60ºC
- Time: 5 - 10 minutes
- Drop the liquor
- Cold wash (2-3 times)

Step 2: Softening

- Water: M:L = 1:6 420 liter
- Machine running
- Flax softener: 0.6 gm/L 252 gm
- Acetic acid: 0.5 gm/L 210 gm (acidic medium help flax softener to work)
- Temperature: Room temperature
- Time: 5 - 10 minutes
- Drop the liquor
- Unload the garments

Step 3: Hydro extraction (700-900 rpm) to remove excess water from the garments for 2-4 minutes.

Step 4: Drying

- Load garments on steam dryer
- Machine running
- Temperature: 60ºC - 70ºC
- Time: 40 - 50 minutes for dry and 10 - 15 minutes for cold dry

- Load garments on gas dryer
- Machine running
- Temperature: 70ºC - 85ºC
- Time: 30 - 35 minutes for dry and 10 - 15 minutes for cold dry

Step 5: QC & Delivery

After drying, quality is checked and good quality garments will be delivered to next section.

Pigment Wash



The wash which is done on pigment to create fading effect is called pigment wash. It is generally done on pigment dyed or printed garments. It is similar to normal wash but a bit costlier. Pigment wash requires a higher temperature of water than normal wash, normally 50-60°C so that the color would fade more evenly.

Objectives

- Create fading/old looking effect on garments and also seam areas.
- Create soft feeling to wear the garment.
- Help to achieve the buyer's washing standard.
- Increase color fastness and rubbing fastness.

Process

Step 1: Pigment Wash

- Lot size: 80 kg
- Water: M:L = 1:8 640 liter
- Machine running
- Caustic soda (NaOH): 0.8 gm/L 512 gm
- Detergent: 0.8 gm/L 512 gm
- Soda ash (Na_2CO_3): 1.5 gm/L 960 gm
- Temperature: 50ºC - 60ºC
- Time: 20 - 60 minutes
- Drop the liquor
- Hot wash for 5 minutes at 50ºC
- Cold wash for 5 minutes

Step 2: Softening

- Water: M:L = 1:6 480 liter
- Machine running
- Flax softener: 0.6 gm/L 288 gm
- Acetic acid: 0.5 gm/L 240 gm (for pH 4 - 5)
- Silicon softener: 0.4 gm/L 192 gm (for softer effect)
- Temperature: Room temperature
- Time: 15 - 25 minutes
- Drop the liquor
- Unload the garments

Step 3: Hydro extraction to remove excess water from the garments for 2-4 minutes.

Step 4: Drying

- Load garments on steam dryer
- Machine running
- Temperature: 60ºC - 70ºC
- Time: 40 - 50 minutes for dry and 10 - 15 minutes for cold dry

Step 5: QC & Delivery

After drying, quality is checked and good quality garments will be delivered to next section.

Caustic Wash



Caustic wash is a chemical washing process used to give garments a light faded or aged appearance. It involves treating garments with a solution of sodium hydroxide (NaOH) and detergent, which breaks down the fibers of the fabric and removes some of the dye. This results in a lighter color and a softer hand feel. Caustic wash is generally done on reactive dyed, sulphur dyed, direct dyed or printed garments.

In case of caustic wash, printing is done on grey fabric without pre-treatment (desizing, scouring, bleaching). These fabrics are used to manufacture garments. Then the garments ate subjected to caustic wash. During caustic wash 20%-30% pigment is washed out, as a result different color fading effect is produced in the garments.

Objectives

- Create little fading/old looking effect on garment.
- Remove size materials from the garments.
- Increase color fastness and rubbing fastness.
- Increase the hairiness on garments.
- Create soft feeling to wear the garments.
- Help to achieve buyer reference sample.

Process

Step 1: Caustic Wash

- Lot size: 80 kg
- Water: M:L = 1:8 640 liter
- Machine running
- Caustic soda (NaOH): 1.6 gm/L 1024 gm
- Detergent: 0.8 gm/L 512 gm
- Temperature: 50ºC - 60ºC
- Time: 20 - 60 minutes
- Drop the liquor
- Cold wash for 3 minutes

Step 2: Neutralization

- Water: M:L = 1:5 400 liter
- Acetic acid: 1 gm/L 400 gm (neutralize garments from alkaline medium)
- Time: 5 minutes

Step 3: Softening

- Water: M:L = 1:6 480 liter
- Machine running
- Flax softener: 0.6 gm/L 288 gm
- Acetic acid: 0.5 gm/L 240 gm
- Silicon softener: 0.4 gm/L 192 gm
- Temperature: Room temperature
- Time: 5 - 10 minutes
- Drop the liquor
- Unload the garments

Step 4: Hydro extraction to remove excess water from the garments for 2-4 minutes.

Step 5: Drying

- Load garments on steam dryer
- Machine running
- Temperature: 60ºC - 70ºC
- Time: 40 - 50 minutes for dry and 10 - 15 minutes for cold dry

Step 6: QC & Delivery

After drying, quality is checked and good quality garments will be delivered to next section.

Limitations

- Process is difficult to control.
- Problem of yellowing.
- Require antichlor treatment which causes corrosion to stainless steel.
- Harmful to human health.
- Source of environmental pollution.

Bleach Wash



Bleach wash is one of the precise garments washing techniques that is used to fade a higher degree of color. It is normally carried out with a strong oxidative bleaching agent such as sodium hypochlorite (NaOCl) or potassium permanganate ($KMnO_4$). Oxidative bleach works by breaking the chemical bonds that make up the chromospheres.

Objectives

- Achieve higher degree of fading effect on garments.
- Remove starch and size materials from garments.
- Create soft feeling to wear the garments.
- Increase color fastness and rubbing fastness.

Process

Step 1: Desizing

- Lot size: 60 kg denim long pant
- Water: M:L = 1:9 540 liter
- Machine running
- Desizing agent: 0.6 gm/L 324 gm
- Detergent / Antistain: 1 gm/L 540 gm
- Temperature: 60ºC
- Time: 15 - 25 minutes
- Drop the liquor

Step 2: Hot Wash

- Water: M:L = 1:9 540 liter
- Temperature: 60ºC
- Time: 5 minutes

Step 3: Bleaching

- Water: M:L = 1:8 480 liter
- Machine running
- Bleaching powder (k.c.i): 10 gm/L 4800 gm
- Soda ash: 5 gm/L 2400 gm
- Temperature: 60ºC
- Time (depend upon the shade): 12 - 15 minutes
- Drop the liquor
- Rinse twice, 3 minutes each

Step 4: Neutralization

- Water: M:L = 1:9 540 liter
- Sodium hyposulphite: 3 gm/L 1620 gm
- Temperature: 40ºC
- Time: 10 - 12 minutes
- Drop the liquor
- Rinse once

Step 5: Softening

- Water: M:L = 1:8 480 liter
- Machine running
- Flax softener: 0.6 gm/L 288 gm
- Acetic acid: 0.5 gm/L 240 gm
- Silicon softener: 0.4 gm/L 192 gm
- Temperature: Room temperature
- Time: 5 - 10 minutes
- Drop the liquor
- Unload the garments

Step 6: Hydro extraction to remove excess water from the garments for 2-4 minutes.

Step 7: Drying

- Load garments on steam/gas dryer
- Machine running
- Temperature 75ºC - 85ºC
- Time: 35 - 45 minutes for dry and 10 - 15 minutes for cold dry

Step 8: QC & Delivery

After drying, quality is checked and good quality garments will be delivered to next section.

Limitations

- The same level of bleaching is very difficult to achieve in repeated runs.
- Bleaching treatment sometimes damages cellulose, resulting in strength loss and or pinholes at the seam, pocket, etc.
- Bleaching liquor is harmful to human health.
- This may also cause corrosion to machine parts.
- Bleaching treatment needs antichlor treatment to eliminate the subsequent yellowness in the fabric.

Peroxide Bleaching & Hypochlorite Bleaching

Hypochlorite as bleaching agents (calcium hypochlorite, sodium hypochlorite etc.) are still popular despite the health and environmental issues. Peroxide is not used in bleaching as it requires 100ºC temperature to work properly. But this high temperature and higher strength of peroxide can damage the garments. Soda ash is also used with peroxide to increase pH (10.5-11.5) and reduce fading strength/speed.

Bleaching with peroxide is costlier than that of hypochlorite. Peroxide bleaching can take longer time than hypochlorite bleaching. Hypochlorite bleach is typically more effective in removing dark colors, achieving high levels of brightness and whitening compared to peroxide bleach. Peroxide bleach has a higher tendency to cause yellowing on certain fabrics if not used properly. Peroxide bleaching can cause catalytic damage to cotton fabrics, which can lead to uneven dyeing or the formation of small holes.

Popularity of Bleach Wash

Bleach can remove natural yellowing in cotton and other fabrics, giving them a brighter, whiter appearance. This is especially important for garments that are intended to be white, such as t-shirts, jeans, and underwear. Bleach is a relatively inexpensive chemical, which makes it a cost-effective way to whiten and finish garments. This is especially important for large-scale garment manufacturers, who need to produce garments quickly and cheaply.

Bleach can be used to achieve a variety of effects on garments, such as softening, brightening, and fading. It can be used on a variety of fabric types, including cotton, denim, and synthetic blends. This makes it a versatile finishing agent that can be used to create a variety of looks. Bleach wash allows for a high level of control and customization. Manufacturers can experiment with different concentrations, application techniques, and exposure times to achieve desired levels of bleaching and create unique effects.

These make bleach wash popular in the apparel industry despite its hazards. Despite its popularity, the use of bleach in industrial settings requires proper handling, safety precautions, and adherence to environmental regulations.

Enzyme Wash



Enzymes are living bio chemical substances that work as catalyst in the washing process. Cellulase enzymes are used in the treatment of denim garment washing. It attacks on indigo color and results in fading effect.

Enzyme mainly hydrolyze the cellulose during wash. At first, it attacks the projecting fibers and hydrolyze them. Then it attacks the yarn portion inside fabric, partly hydrolyze the portion and produce faded effect.

Objectives

- Achieve required fading effect as per sample.
- Remove starch and size materials from the garments.
- Create soft feeling to wear the garment (enzyme loosen the structure of fabric).
- Increase the color fastness and rubbing fastness.
- Develop “Bio-Polishing” effect or smooth surface on cotton / denim.

Classifications

A. Acid Enzyme (liquid)

- Color is slightly brown.
- pH range is 4.5 - 5.5.
- Temperature: 40ºC - 55ºC.
- Time: 25 - 50 minutes.
- Enzyme effects come within fast.
- More staining / bleeding on garments.

B. Neutral Enzyme

i) Powder Form

- Slightly white powder form.
- pH range is 6 - 7.
- Temperature: 40ºC - 60ºC.
- Time: 40 - 70 minutes.
- Enzyme effects come slowly.
- Less staining / bleeding on garments.
- Good abrasion effect on garments.
- Good effect in dark shade.

ii) Liquid Form (SL Enzyme)

- SL enzyme is liquid form.
- pH range is 6 - 7.
- Temperature: 40ºC - 60ºC.
- Time: 45 - 80 minutes.
- Enzyme effects come slowly.
- Less staining / bleeding on garments.
- Better abrasion effect on garments.
- Good effect in dark shade.

Control Parameters

- Temperature

Temperature control is important to achieve desired effect. Higher temperatures can enhance enzyme activity, but there is a limit beyond which the enzymes may lose effectiveness. Most enzymes work best at a temperature of around 40ºC to 60ºC.

- pH

Enzymes have an optimal pH range in which they work most efficiently. Acidic enzyme works best in 4.5 - 5.5 pH range. Neutral enzyme works best in 6 - 7 pH range.

- Time

Longer reaction times will produce a more pronounced effect, but they may also damage the fabric if they are too long. Normally 40 to 70 minutes washing time required according to shade.

- Enzyme Type

Type of enzyme used will affect the overall properties of the finished fabric. A neutral enzyme will produce a softer hand feel, while an alkaline enzyme will produce a more crease-resistant fabric.

Process

Step 1: Desizing

- Lot size: 60 kg denim long pant
- Water: M:L = 1:9 — 540 liter
- Machine running
- Desizing agent: 0.6 gm/L — 324 gm
- Detergent: 0.8 gm/L — 432 gm
- Temperature: 60ºC
- Time: 15 - 25 minutes
- Drop the liquor
- Cold wash once

Step 2: Enzyme Wash

- Water: M:L = 1:8 480 liter
- Temperature: 45ºC
- Acetic acid: 0.6 gm/L 288 gm
- Anti back staining: 0.6 gm/L 288 gm
- Acid enzyme: 2 gm/L 960 gm
- Time: 40 - 50 minutes
- Increase temperature to 90ºC and run 1 minute (enzyme killing)
- Drop the liquor
- Rinse twice, 3 minutes each

Step 4: Neutralization

- Water: M:L = 1:9 540 liter
- Sodium metabisulphite: 3 gm/L 1620 gm
- Temperature: 40ºC
- Time: 10 - 12 minutes
- Drop the liquor
- Rinse once

Step 5: Softening

- Water: M:L = 1:8 480 liter
- Acetic acid: 0.6 gm/L 288 gm
- Cationic softener: 1 gm/L 480 gm
- Temperature: Room temperature
- Time: 15 - 20 minutes
- Drop the liquor
- Unload the garments

Step 6: Hydro extraction to remove excess water from the garments for 2-4 minutes.

Step 7: Drying

- Load garments on steam/gas dryer
- Machine running
- Temperature 75ºC - 85ºC
- Time: 35 - 45 minutes for dry and 10 - 15 minutes for cold dry

Step 8: QC & Delivery

After drying, quality is checked and good quality garments will be delivered to next section.

Advantages *(why it is popular)*

- Environment friendly wash.
- Softer hand feel.
- Reduced damage to fabric.
- Improves luster and color fastness.
- Prevents the tendency of pilling after a relatively short period of wear.
- Less time is needed at rinsing after completing enzyme washing than stone washing.
- Most economical due to more water saving.
- Reduce labor engagement.
- The enzymes can be recycled.

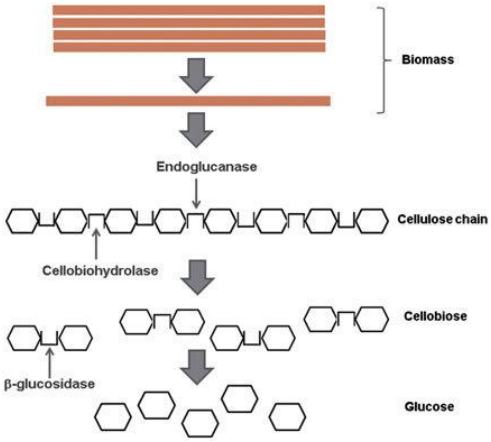
Disadvantages

- Chances of back staining in the garments.
- Post-treatment required.
- To remove the back staining, the garments are rigorously washed.
- Enzymes can be relatively expensive compared to traditional wash chemicals.
- Enzyme wash is not suitable for all fabrics. Silk and wool can be damaged by enzymes.
- Enzyme wash can damage fabric if it is not done properly.

Mechanism & Reaction of Cellulase Enzyme

Cellulase enzymes are produced by certain microorganisms and are capable of hydrolyzing cellulose. The mechanism of how cellulase enzymes hydrolyze cellulose involves several steps. Cellulase enzymes first adsorb onto the surface of the cellulose fiber. This step involves the binding of the enzyme to specific sites on the cellulose surface.

Once adsorbed, the cellulase enzyme initiates the hydrolysis of cellulose through the action of its different components:



Endoglucanase: These enzymes randomly attack the amorphous regions of cellulose chains, breaking them into smaller fragments.

Exoglucanase: These enzymes act on the ends of cellulose chains, cutting off disaccharide units from the non-reducing ends.

β-glucosidase: These enzymes hydrolyze the disaccharide units released by endoglucanase and exoglucanase into individual glucose molecules.

The reaction involved in cellulase hydrolysis of cellulose is the enzymatic cutting of the β-1,4-glycosidic bonds that link the glucose units in cellulose. This reaction results in the breakdown of cellulose into smaller fragments and ultimately glucose. The overall reaction can be represented as:

Cellulose + H_2O → Glucose

The efficiency of cellulase hydrolysis depends on factors such as enzyme concentration, pH, temperature, and substrate accessibility. Optimal conditions need to be maintained to ensure effective hydrolysis of cellulose by cellulase enzymes.

Why enzyme wash is sustainable wash?

Enzyme wash offers a more sustainable alternative to traditional washing methods by reducing chemical usage, conserving water and energy, minimizing waste, and promoting garment longevity. These factors contribute to its reputation as a more environmentally friendly choice in the apparel industry.

Reduced Chemical Usage: Enzymes used in the wash process are derived from natural sources and are biodegradable. They help break down and remove unwanted substances from the fabric, reducing the need for harsh chemicals such as acids or alkalis.

Energy Efficiency: Enzyme wash typically operates at lower temperatures, which reduces the energy required for heating the water. Lower energy consumption translates to reduced greenhouse gas emissions and contributes to overall energy efficiency.

Water Conservation: Enzyme wash requires less water compared to traditional washing methods. This conserves water resources and reduces the strain on water supply systems.

Waste Reduction: Enzyme wash can be targeted to specific areas of the garment, reducing the need for overall washing and minimizing the production of wastewater. This helps in waste reduction and decreases the environmental impact associated with wastewater treatment.

Extended Garment Lifespan: Enzyme wash is gentle on fabrics and helps to maintain their integrity. By removing surface impurities without causing excessive wear or damage, enzyme wash can extend the lifespan of garments.

How enzyme wash creates softer hand feel

Enzyme wash creates softer hand feel by breaking down the bonds in the fabric, making it more pliable and comfortable to wear. Enzymes are used to break down the cellulose fibers in the fabric. One commonly used enzyme, cellulase, targets and breaks down cellulose fibers in cotton or cellulose-based fabrics. This hydrolysis action removes small fibrils and surface protrusions, resulting in a smoother texture and softer feel. Enzymes also modify the surface structure of the fabric, reducing fiber stiffness and enhancing flexibility.

Why neutralization step is necessary in acid enzyme wash?

In acid enzyme wash, enzymes are generally deactivated simply by raising the temperature. Sodium metabisulphite is used in the neutralization step of acid enzyme wash to neutralize the acid and to prevent the enzymes from reactivating.

Enzymes are proteins that are deactivated by heat, meaning that they are changed in their structure and become inactive. However, enzymes can be reactivated if they are exposed to a proper medium. Sodium metabisulphite is a reducing agent that reacts with the enzymes and prevents them from reactivating. It also helps to neutralize the acid, which is important because acids can damage fabric and cause skin irritation.

Problems of Extra Enzyme Wash

While enzyme wash is a popular technique in garment washing, there can be some problems associated with an excessively long enzyme washing process. These problems include:

Over Fading: Prolonged exposure to enzymes can lead to excessive fading of the fabric, resulting in an uneven or patchy appearance.

Fabric Weakening: Extended enzyme treatment can weaken the fabric structure, especially if the enzymes are too concentrated or the process is carried out at high temperatures.

Loss of Dimensional Stability: Excessive enzyme washing can cause the fabric to lose its dimensional stability, resulting in shrinkage or distortion of the garment. Also, the fabric can become too soft or too rough.

Increased Chemical Consumption: Longer enzyme washing processes require more enzymes and chemicals, leading to higher consumption and potentially increasing the environmental impact.

Conducting trials and testing on sample garments can help determine the optimal enzyme treatment duration to achieve the desired wash effect without compromising the quality and integrity of the fabric.

Enzyme Wash & Bleach Wash

Bleach can produce very light fading effect than that of enzyme. Excess use of enzyme can reduce fabric strength. Any type of fading effect can be achieved using bleach wash. Additionally, bleach is cheaper than enzyme. That's why instead of being environmentally hazardous, it is used in industry.

Here are some differences between enzyme wash and bleach wash-

Enzyme Wash	Bleach Wash
Achieve specific effects (softening, fading, distressing)	Alter the color of the fabric (lightening or bleaching)
Enzymes break down and remove surface colors from fibers	Bleaching agents oxidize or remove dye molecules
Softens fabric, creates texture, specific fading effects	Alters color, creates lightened or bleached appearance
Gentle in nature	More aggressive in nature
More environmentally friendly	Negative impact if not properly managed
Generally safer to handle	Chemicals can be hazardous if mishandled
Less likely to damage fabric	More likely to damage fabric

Stone Wash

Pumice stone is a volcanic rock that consists of highly vesicular or perforated volcanic glass. It is created when super-heated and highly pressurized rock is violently erupted from a volcano. The unusual foamy configuration of pumice stone happens because of simultaneous rapid cooling and rapid de-pressurization. It can be added with any wash to achieve irregular 'used' or 'vintage' look on the garments.

Characteristics of Pumice Stone

- Perforated stones, produced from volcanic explosion.
- Have soaking ability due to perforated surface.
- Float on water.
- At first the stones are soft but later becomes hard with rough surface.
- Available in Indonesia, Turkey. Indonesian stone is slightly brown, Turkish stone is white.
- Available in 3 sizes. Small 2-3 cm, medium 4-5 cm and large size 6-7 cm.
- Hardness in Mohr scale is 6.
- Melting temperature is 15000ºC.
- pH value is 8.
- Density is 2.313 gm/cm^3.
- Same stones can be used 2-3 times.

Chemical Composition of Pumice Stone

- Silicon Dioxide, SiO_2 - 71.91%
- Aluminum Oxide, Al_2O_3 - 12.66%
- Potassium Oxide, K_2O - 4.30%
- Sodium Oxide, Na_2O - 3.45%
- Calcium Oxide, CaO - 1.46%
- Ferric Oxide, Fe_2O_3 - 1.13%
- Magnesium Oxide, MgO - 0.32%
- Sulphur Trioxide SO_3 - 0.03%
- Calcification Loss - 4.53%
- Indefinable Content - 0.21%

Working Principle of Pumice Stone

Pumice stones are used to create irregular fading effect on garments. They perform brushing action on the garment fabric surfaces. The areas where more brushing action takes place, more fading or discolor effect will be developed there and the areas where less brushing action takes place, less discolor effect will be developed there. The multi-layer fabric areas like- collar, cuff, pocket, placket, side seams etc. areas will be brushed more than the single layer areas.

To get the desired washed effect, the stones should be of proper hardness, shape and size. Degree of color fading depends on some factors, such as- the garments to stone ratio, washing time, stone size, and material to liquor ratio. Stone washing times vary from 60-120 min. Stones can be reused until they disintegrate completely. Small stones can provide an even and uniform abrasion effect, but the color contrast produced is not good when compared with large stones.

Advantages of Pumice Stone

- Desired vintage effect on garments can be easily achieved by using pumice stones.
- Washed garments can achieve more soft effect by using pumice stones.

Disadvantages of Pumice Stone

- Can damage washing machine due to stone and machine abrasion.
- Garments can be damaged due to stone and machine abrasion.
- Creates environmental problem of disposition of waste of the grit produced by the stones.
- High labor costs due to labor intensive processes.
- Several washes require in order to completely get rid of the stones.

Stone Bleach Wash



Stone washing of denim fabric gives irregular ‘used’ or ‘vintage’ look on the garments. This is due to the varying degree of abrasion in the garment which scrap off dye particles from the surface of the yarn. Addition of pumice stone in bleach wash accelerates mechanical abrasion and allow irregular faster degradation of surface color.

Objectives

- Create irregular fading effect on garments.
- Remove the size materials from the garments.
- Remove dust, dart, oil spot, impurities from the garments.
- Create soft feeling to wear the garments.

Process

Step 1: Desizing

- Lot size: 60 kg denim long pant
- Water: M:L = 1:9 — 540 liter
- Machine running
- Temperature: 60ºC
- Desizing agent: 0.6 gm/L — 324 gm
- Detergent / Antistain: 1 gm/L — 540 gm
- Time: 15 - 25 minutes
- Drop the liquor

Step 2: Hot Wash

- Water: M:L = 1:9 540 liter
- Temperature: 60ºC
- Time: 5 minutes

Step 3: Stone Bleaching

- Water: M:L = 1:8 480 liter
- Machine running
- Bleaching powder (k.c.i): 10 gm/L 4800 gm
- Soda ash: 5 gm/L 2400 gm
- Pumice stone: ½ the weight of garments
- Temperature: 60ºC
- Time: 12 - 15 minutes
- Drop the liquor
- Rinse twice, 3 minutes each
- Pumice stone out

Step 4: Neutralization

- Water: M:L = 1:9 540 liter
- Sodium hyposulphite: 3 gm/L 1620 gm
- Temperature: 40ºC
- Time: 10 - 12 minutes
- Drop the liquor
- Rinse once

Step 5: Softening

- Water: M:L = 1:8 480 liter
- Machine running
- Cationic softener: 1 gm/L 480 gm
- Acetic acid: 0.6 gm/L 288 gm
- Temperature: Room temperature
- Time: 5 - 10 minutes
- Drop the liquor
- Unload the garments

Step 6: Hydro extraction to remove excess water from the garments for 2-4 minutes.

Step 7: Drying

Step 8: QC & Delivery

Stone Enzyme Wash

Stone enzyme wash means washing the garments with special pumice stones along with enzymes. The stones perform mechanical rubbing action and enzyme perform chemical action which altogether create a strong and unique faded look on garments.



Objectives

- Create irregular fading effect on garments.
- Remove the size materials from the garments.
- Remove dust, dart, oil spot, impurities from the garments.
- Create soft feeling to wear the garments.

Process

Step 1: Desizing

- Lot size: 60 kg denim long pant
- Water: M:L = 1:9 540 liter
- Machine running
- Temperature: 60ºC
- Desizing agent: 0.6 gm/L 324 gm
- Detergent / Antistain: 1 gm/L 540 gm
- Time: 15 - 25 minutes
- Drop the liquor

Step 2: Hot Wash

- Water: M:L = 1:9 540 liter
- Temperature: 60ºC
- Time: 5 minutes

Step 3: Stone Enzyme Wash

- Water: M:L = 1:8 480 liter
- Pumice stone: ½ the weight of garments
- Acetic acid: 0.6 gm/L 288 gm
- Anti back staining: 0.8 gm/L 384 gm
- Acid enzyme: 1.5 gm/L 720 gm
- Temperature: 40ºC to 50ºC
- Time: 60 - 70 minutes
- Increase temperature to 90ºC and run 1 minute (enzyme killing)
- Drop the liquor
- Rinse twice, 3 minutes each
- Pumice stone out

Step 4: Neutralization

- Water: M:L = 1:9 540 liter
- Sodium metabisulphite: 3 gm/L 1620 gm
- Temperature: 40ºC
- Time: 10 to 12 minutes
- Drop the liquor
- Rinse once

Step 5: Softening

- Water: M:L = 1:8 480 liter
- Acetic acid: 0.6 gm/L 288 gm
- Cationic softener: 1 gm/L 480 gm
- Temperature: Room temperature
- Time: 5 minutes
- Drop the liquor
- Unload the garments

Step 6: Hydro extraction to remove excess water from the garments for 2-4 minutes.
Step 7: Drying
Step 8: QC & Delivery

Acid Wash



Acid wash is a process in which a garment is treated with oxidizing agent soaked stones to produce irregular fading effect. Brushing action of pumice stones creates the irregular fading effect on garments. The acid used is typically a weak solution of hydrochloric acid or a similar weak acid. Results of the effect depend upon dyestuff, fabric, concentration of chemicals, stones. It is normally done on the garments from heavy fabrics like denim, course canvas, sweater and thick twill etc.

The wash is performed in two steps- garments are washed without water in first step, and with water in second step. Pumice stones are soaked in a strong oxidizing agent such as sodium hypochlorite or potassium permanganate. Stones absorb chemicals and become saturated. Excess liquor is removed and the stones are dried in normal air or sun.

Desized dried garments are put in a tumble washer filled with treated stones. Tumble washer is run to wash the garments without water. Stones abrade the garments, especially, the exposed parts. Then garments are taken out and transferred to another tumbler filled with water for washing and rinsing. After rinsing, prominent acid wash effect shows up. Whitening agents are often added during rinse to make the effect more prominent.

Objectives

- Produce irregular fading effects on the garments surface.
- Create vintage look.
- Improve softness.
- Achieve the buyer washing standard.
- Increase rubbing fastness.

Process

Step 1: Desizing

- Lot size: 60 kg denim pant
- Water: M:L = 1:10 600 liter
- Machine running
- Desizing agent: 1 gm/L 600 gm
- Detergent: 1 gm/L 600 gm
- Temperature: 60ºC
- Time: 20 minutes
- Drop the liquor
- Cold rinse once for 3 minutes

Step 2: Hot Wash *(to remove the adhering materials)*

- Water: M:L = 1:10 600 liter
- Temperature: 60ºC
- Time: 5 minutes
- Drop the liquor
- Unload the garments
- Load the pre-treated garments in dryer
- Dry the garment completely & unload

Step 3: Acid Wash

Preparation of Pumice Stone

- Water: 100 liter
- Potassium per manganate: 1000 gm
- Phosphoric acid: 250 gm
- Stir the solution in a stainless steel tub with dry pumice stone
- Soak the stones with chemical solution for 10 - 15 minutes
- Dry the soaked stones in open air for 2 to 3 hours



Dry Wash

- Load pre-treated garments (30-40 kg per batch) in dry washing machine (without water)
- Load per-treated stones: ½ the weight of garments
- Machine running
- Time: 7 - 10 minutes (for each batch)
- Unload treated garment
- Load stone treated garments in another washing machine

Step 4: Washing

- Lot size: 60 kg
- Water: M:L = 1:8 480 liter
- Detergent: 1 gm/L 480 gm (remove stone dust and chemicals)
- Temperature: 40ºC - 50ºC
- Time: 10 minutes
- Drop the liquor

Step 5: Neutralization *(whitening)*

- Water: M:L = 1:8 480 liter
- Sodium metabisulphite: 5 gm/L 2400 gm
- Temperature: Cold
- Time: 5 minutes
- Drop the liquor

Step 6: Softening

- Water: M:L = 1:7 420 liter
- Acetic acid: 0.6 gm/L 252 gm
- Softener: 1 gm/L 420 gm
- Temperature: Room temperature
- Time: 5 minutes
- Drop the liquor
- Unload the garments

Step 7: Hydro extraction to remove excess water from the garments for 2-4 minutes.

Step 8: Drying

Step 9: QC & Delivery

After drying, quality is checked and good quality garments will be delivered to next section.

Precautions

- Operator should wear protective equipment and safety glasses.
- Have to maintain stock solution and stock solution consistency during socking the stones.
- No water should leakage into the machine during stone and garments treatment in machine.
- Have to add some stone after finishing first batch to maintain the volume for second batch.
- Proper neutralization and rinsing to remove yellowness due to manganese.

Why soaked stone is used in acid wash?

Soaked stones are used in acid wash because they help to create a more localized and uneven fading effect. The pumice stones are naturally porous, which allows them to absorb the strong oxidizing agent used in the process. This means that the dye in the denim is more likely to fade where the stones come into contact with the fabric. This results in a more distressed and vintage look.

In addition, the pumice stones help to abrade the surface of the fabric, which also contributes to the fading effect. The stones act like a mild abrasive, breaking down the fibers of the fabric and making them more pliable. This also helps to remove any excess dye from the fabric, resulting in a lighter overall color. If the stones were not soaked, they would not be able to absorb the oxidizing agent as effectively. This would result in a more uniform fading effect, which is not typically desired in an acid wash.

Acid Wash of Knit Garments

Thermocol balls are used instead of pumice stone on acid wash of knitted garments. Thermocol balls are made of a lightweight, flexible material that is able to abrade the surface of the fabric without damaging it and help to create a more uneven and distressed look.

Thermocol balls are lighter and more flexible than pumice stones. This makes them easier to handle and less likely to damage the fabric. They are less likely to leave sharp edges on the fabric. This is important because sharp edges can cause the fabric to fray or pill. They are more uniform in size than pumice stones. This helps to ensure that the acid is distributed evenly throughout the garment.

Acid Wash & Bleach Wash

Acid wash uses a strong oxidizing agent, such as potassium permanganate, to fade the dye in the fabric. On the other hand, bleach wash uses a strong bleach, such as sodium hypochlorite, to fade the dye in the fabric.

Acid Wash	Bleach Wash
Potassium permanganate is used	Sodium hypochlorite is used
Uneven and distressed effect	Uniform and washed-out effect
Softer and more pliable feeling	Stiffer and more brittle feeling
Reduce durability of fabric	Doesn’t reduce durability of fabric compared to acid wash

Silicon Wash

Silicon wash is a finishing process that is used to give fabrics a soft, slippery, and durable feel. The silicone-based chemicals create a thin, invisible film on the fabric surface, reducing friction between fibers and providing a silky touch. This process also helps to reduce fabric shrinkage and improve the overall appearance of the garment. It is typically used on denim, but can also be used on other fabrics, such as twill, canvas, knit, corduroy etc.



Objectives *(advantages)*

- Gives elastic handle.
- Gives durable softness.
- Helps to tear resistance, anti-pilling effects, and dimensional stability.
- Helps to fabrics to be cut and sewn more easily.
- Improve wear and easy care properties.

Process

Step 1: Desizing

- Lot size: 60 kg twill/canvas/poplin/corduroy trouser
- Water: M:L = 1:10 600 liter
- Machine running
- Desizing agent: 0.6 gm/L 360 gm
- Detergent: 0.5 gm/L 300 gm
- Temperature: 50ºC
- Time: 10 - 20 minutes
- Drop the liquor
- Rinse once for 3 minutes

Step 2: Silicon Wash

- Water: M:L = 1:8 480 liter
- Acetic acid: 0.6 gm/L 288 gm
- Cationic softener: 1 gm/L 480 gm
- Silicon (ME): 0.5 gm/L 240 gm
- Temperature: 40ºC
- Time: 15 - 20 minutes
- Drop the liquor
- Unload the garments

Step 3: Hydro extraction to remove excess water from the garments for 2-4 minutes.

Step 4: Drying

- Load garments on steam/gas dryer
- Machine running
- Temperature 75ºC - 85ºC
- Time: 35 - 45 minutes for dry and 10 - 15 minutes for cold dry

Step 5: QC & Delivery

After drying, quality is checked and good quality garments will be delivered to next section.

Limitations

- In some cases, silicon wash can make fabrics feel stiff or plasticky.
- Silicon film on the fabric surface can slightly reduce the breathability of the garment.
- Some individuals may be allergic to the silicone-based chemicals used in the process.
- Disposal of silicone-based chemicals can have environmental implications.
- Silicon wash may cause slight color alteration or loss of vibrancy, particularly in bright or highly saturated colors.

Super White Wash



Super white wash is a specialized washing technique used to achieve a bright, crisp white appearance on garments. Super white wash is done on the garments made from grey fabrics. It is typically used on cotton fabrics, but can also be used on other fabrics, such as linen and polyester.

The process involves the use of optical brighteners or fluorescent whitening agents, which are chemicals that absorb ultraviolet light and emit blue light, thus enhancing the perceived whiteness of the fabric. These agents work by absorbing invisible ultraviolet light and re-emitting it as visible blue light, which counteracts any yellowish or dull tones in the fabric, resulting in a brighter white appearance.

Objectives *(advantages)*

- Make whiter garments.
- Create soft feeling to wear the garments.
- Remove size materials from the garments.
- Improve comfortability of the garments.
- Change the outlook of the garments.
- Help to achieve buyer required washing standard.

Process

Step 1: Desizing

- Lot size: 60 kg cotton shirt (grey fabrics)
- Water: M:L = 1:8 480 liter
- Machine running
- Temperature: 90ºC
- Detergent: 2% owf 1.2 kg
- Caustic soda: 8% owf 4.8 kg
- Soda ash: 5% owf 3 kg
- Hydrogen peroxide: 12% owf 7.2 kg
- Stabilizer: 5% owf 3 kg
- Time: 70 minutes
- Drop the liquor

Step 2: Bleaching

- Water: M:L = 1:8 480 liter
- Detergent: 2% owf 1.2 kg
- Caustic soda: 4% owf 2.4 kg
- Hydrogen peroxide: 10% owf 6 kg
- Stabilizer: 2% owf 1.2 kg
- Temperature: 90ºC
- Time: 60 - 90 minutes
- Drop the liquor

Step 3: Hot Wash

- Water: M:L = 1:8 480 liter
- Temperature: 50ºC
- Time: 5 minutes

Step 4: Neutralization

- Water: M:L = 1:8 480 liter
- Acetic acid: 1% owf 600 gm
- Time: 5 minutes
- Drop the liquor

Step 5: Brightening

- Water: M:L = 1:7 420 liter
- Optical brightener agent: 0.5% owf 300 gm
- Time: 5 minutes
- Drop the liquor

Step 6: Softening

- Water: M:L = 1:7 420 liter
- Non-ionic softener: 0.4% owf 240 gm
- Micro emulsion silicon: 0.4% owf 240 gm
- Time: 5 minutes
- Drop the liquor
- Unload the garments

Step 7: Hydro extraction to remove excess water from the garments for 2-4 minutes.

Step 8: Drying

Step 9: QC & Delivery

Limitations

- Dependency on Optical Brighteners

The effectiveness of super white wash relies on the optical brighteners or fluorescent whitening agents, which may have limitations in certain lighting conditions or over time. The whiteness achieved through this process may gradually fade with repeated washing and exposure to sunlight.

- Environmental Considerations

Some optical brighteners used in super white wash may have environmental implications if not properly managed or disposed of. Eco-friendly practices and compliance with regulations are necessary to minimize any negative environmental impact.

Why bleaching is done twice and H_2O_2 is used?

Bleaching is mainly used for decoloring and natural color removing is done better with the help of H_2O_2, hydrogen peroxide. That's why hypochlorite bleach is not used. Also bleaching is done twice in the super white wash to achieve a high level of whiteness and brightness in the fabric.

The first bleaching removes any natural yellowing or discoloration in the fabric. The second bleaching further brightens the fabric and helps to remove any remaining impurities. The repeated bleaching also allows for better control over the level of whiteness, as it helps to ensure that all areas of the fabric are evenly treated. This helps to achieve a consistent and uniform white color throughout the fabric, enhancing its overall aesthetic appeal. The volume is reduced to about 50% for second bleaching to reduce the strength loss of fabric due to repeated bleaching.

Topics	***Normal***	***Pigment***	***Caustic***	***Bleach***	***Enzyme***	***Acid***	***Silicon***
Desizing				✓	✓	✓	✓
Cold Wash					✓	✓	✓
Hot Wash				✓		✓	
Specific Wash	M:L = 1:8 Detergent: 0.8 gm/L 50°C - 60°C 5 - 10 min	M:L = 1:8 Caustic soda: 0.8 gm/L Detergent: 0.8 gm/L Soda ash: 1.5 gm/L 50°C - 60°C 20 - 60 min * Hot wash 5 min 50°C	M:L = 1:8 Caustic soda: 1.6 gm/L Detergent: 0.8 gm/L 50°C - 60°C 20 - 60 min	M:L = 1:8 Bleaching powder: 10 gm/L Soda ash: 5 gm/L 60°C 12 - 15 min	M:L = 1:8 45°C Acetic acid: 0.6 gm/L Anti back staining: 0.6 gm/L Acid enzyme: 2 gm/L 40 - 50 min 90°C kill enzyme	Water: 100 liter $KMnO_4$: 1000 gm Phosphoric acid: 250 gm Soak 10 - 15 min Dry 2 to 3 hour Load w/o water Soaked stone 7 - 10 min	M:L = 1:6 Flax softener: 0.6 gm/L Acetic acid: 0.5 gm/L Silicon softener: 0.4 gm/L 40°C 5 - 10 min
Cold Wash	✓	✓	✓	✓	✓	M:L = 1:8 Detergent: 1 gm/L 40°C - 50°C 10 min	
Neutralizing			M:L = 1:8 Acetic acid: 1 gm/L 5 min	M:L = 1:8 Na hyposulphite: 3 gm/L 40°C 10 - 12 min	M:L = 1:8 Na metabisulphite: 3 gm/L 40°C 10 - 12 min	M:L = 1:8 Na metabisulphite: 3 gm/L Cold temperature 5 min	
Softening	✓	✓	✓	✓	✓	✓	Done in Specific Wash
Hydroextract	✓	✓	✓	✓	✓	✓	✓
Drying	✓	✓	✓	✓	✓	✓	✓

Desizing

- M:L = 1:8
- Desizing agent: 0.6 gm/L
- Detergent: 0.8 gm/L
- 60°C
- 15 - 25 min

Cold Wash

- Rinse once, 3 min

Hot Wash

- M:L = 1:9
- 60°C
- 5 min

Specific Wash

Cold Wash (Rinse)

- 2-3 times, 3 min

Softening

- M:L = 1:6
- Flax softener: 0.6 gm/L
- Acetic acid: 0.5 gm/L
- Silicon softener: 0.4 gm/L
- Room temperature
- 5 - 10 min

Drying

- Loading
- 75°C - 85°C
- 35 - 45 min dry
- 10 - 15 min cold dry

Topics	***Super White***
Desizing	M:L = 1:8 Detergent: 2% owf Caustic soda: 8% owf Soda ash: 5% owf Hydrogen peroxide: 12% owf Stabilizer: 5% owf 90ºC 70 min
Bleaching	M:L = 1:8 Detergent: 2% owf Caustic soda: 4% owf Hydrogen peroxide: 10% owf Stabilizer: 2% owf 90ºC 60 - 90 min
Hot Wash	M:L = 1:9 60ºC 5 min
Neutralizing	M:L = 1:8 Acetic acid: 1% owf 5 min
Brightening	M:L = 1:7 Optical brightener agent: 0.5% owf 5 min
Softening	M:L = 1:6 Non-ionic softener: 0.4% owf Micro emulsion silicon: 0.4% owf 5 min
Hydroextract	✓
Drying	✓

Dry Process

The finishing process which is applied at dry condition of the garments is called the dry process. It changes the visual appearance of garments by mechanical abrasion without altering the construction and properties. This process is generally done at the unwashed condition of the garments. Apparel gets a nice look after dry process and also it adds value to the product. Most of the time dry process is done manually.

PP Spray

PP spray, also known as potassium permanganate spray, is a chemical treatment used in garment washing to create an old or vintage look on denim or other fabrics. Due to this process, local abraded area appears whiter than background indigo color shade. This process is very complicated and needs highly skilled operators to execute it followed by immediate neutralization. It is often done on denim, twill, poplin, corduroy fabrics and it is done after the first wash.

PP spray process involves spraying a solution of potassium permanganate onto the fabric, typically with the use of a spray gun or machine. The solution reacts with the fabric, causing it to bleach or fade in specific areas, creating a unique pattern or effect. PP spray is often used in combination with other techniques such as sandblasting, whiskering, or scraping to create a more pronounced distressed look on the fabric.

Process

At first garments are desized and fitted in a suitable air dummy. Then potassium permanganate is sprayed in specific parts of the garment to fade color and bring a bright effect on that area. Liquid potassium permanganate color is pink, but after applied to garment it turns into muddy brown or grey on drying. Sometimes acetic acid or phosphoric acid is mixed to increase its intensity. It is always followed by the neutralization by sodium meta bi sulphate or other neutralizing agents. The concentration of potassium permanganate controls the extent of brightness or effect.

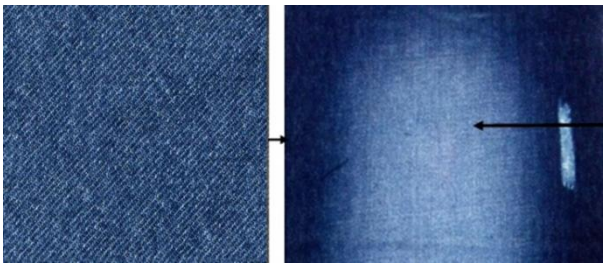
Stock Solution Preparation

- Water: 1 Liter
- Potassium Permanganate: 0.5 - 15 gm/L (standard stock solution = 0.4%)
- Acetic Acid: As per needed

Indigo dyed fabrics treated with low concentration, whereas sulphur black dyed fabrics need high concentration to treat.

Apparatus

- PP spray cabin
- Spray gun
- Air dummy
- Water circulation system
- Musk, gloves, safety shoes & apron



Controlling Parameters

Distance: Less distance will give more defined and sharp effect and more distant will give less bright and mild effect.

Air to Water Ratio of Gun: Low air pressure will result in dropping the solution on the garments instead of spraying which will result in bright spot.

Concentration: This will control the extent of fading.

PP Sponging

PP sponging is a garment washing process that involves using a sponge or other applicator to selectively apply a solution of potassium permanganate (PP) to specific areas of the garment, creating a faded or distressed effect by rubbing. PP sponging is often used in combination with other garment washing techniques, such as enzyme washing, stone washing, or bleach washing, to create a variety of effects on the fabric.

Process



The garment is first washed and dried to remove any dirt, debris, or sizing agents that may be present. A solution of 20% potassium permanganate is prepared by dissolving the chemical in 80% water. Then garment is placed with rubber board on a flat table and the PP solution is applied using a sponge or other applicator. The solution is applied in a controlled manner to create the desired effect, with more solution applied in areas where a more pronounced effect is desired.

After the desired effect is achieved, the garment is neutralized by applying a solution of hydrogen peroxide or sodium bisulfite. This helps to stop the chemical reaction and prevent further bleaching or fading. The garment is then rinsed thoroughly to remove any remaining chemical residue, and then dried.

PP Spray & PP Sponging

The main difference between PP spray and PP sponging is the method of application of the potassium permanganate (PP) solution on the garment.

PP spray involves spraying the PP solution onto the garment using a spray gun or machine. The solution is applied in a controlled manner to create a faded or distressed effect on the entire garment or specific areas of the garment. PP spray can create a variety of effects, including fading, bleaching, or color variation, depending on the concentration of the solution and the method of application.

PP sponging, on the other hand, involves using a sponge or other applicator to selectively apply the PP solution to specific areas of the garment. The solution is applied in a controlled manner to create a faded or distressed effect only in the targeted areas. PP sponging is typically used for creating localized effects such as faded patches, whiskering, or moustaches on denim garments.

Sand Blasting

Sand blasting is a mechanical dry washing process by which Al_2O_3 is passed through a nozzle with the help of high-speed compressed air and pressure is applied onto specific areas of the garment. Sand blasting technique involves removing the dark indigo pigmentation from the apparel propelling a stream of abrasive material (sand) against the fabric under high pressure, gradually softens and lightens the denim. The desired finishing result can be achieved by increasing or decreasing the exposure. Basically, sandblasting is used in heavy garments like- denim, twill, canvas, corduroy etc.

Process

The garments are placed on the bed of a closed chamber. Aluminum oxide (Al_2O_3) is used for sand blasting which looks like sand. Before Al_2O_3, Silicon di Oxide (SiO_2) was used. Aluminum oxides are very fine and hard/sharp. They are thrown by a gun with dry air to perform sand blasting action. The garments are instantly faded by the frictional effect of aluminum oxide. The flow of aluminum oxide on fabric surface is done at 10-20º angles (blowing angle). Higher the blowing angle higher the fading effect and higher the risk of fabric damage. Sand blasting requires a minimum pressure of 60-70 PSI and ideal pressure is 90-100 PSI. Blasting time depends on fading area and depth of fading.

Advantages



- It is a purely mechanical process that uses no chemicals.
- It is a water-free process; therefore, no drying is required.
- A variety of distressed or abraded looks is possible.
- Any number of designs can be created by special techniques.

Disadvantages

- Sand blasting causes silicosis that's why it was banned by some brands.
- Not environment-friendly.
- There is a chance of fabric damage.

Sand Blasting Gun

- Body: Plastic ABS
- Air Inlet: 1/4"
- Operating Pressure: 90 PSI
- Air Flow: 1.41 CFM at 100 PSI
- Nozzle Size: Length 2-15 inch, Dia 7/32 inch
- Air Consumption: 7S CFM
- Storage Case: 600 CC

Sand Blasting for Metal

Sandblasting is the process of smoothing and cleaning a hard surface by forcing solid particles across that surface at high speeds using compressed air. Sandblasting can remove paint, rust, and residue of oxidation from materials quickly and efficiently. Sandblasting can also be used to change the condition of a metal's surface, such as through removing scratches or casting marks.

There are two kinds of sandblasting system- 1. "Suction" or "Siphon" 2. "Pressure" Blast Systems.

There are two basic kinds of sand blasters- 1. Blast Cabinets 2. Portable Blasters.

A sandblasting system includes four basic components-

- Air source
- Sandblasting cabinet
- Dust collector
- Blasting media

The air source is usually bottling gas or an air compressor. The sandblasting cabinet holds the object being blasted, while the dust collector removes dust from the cabinet. The materials used in sandblasting, whether sand or not, are referred to as blast media.

Suction System in Blast Cabinet

- Open door
- Place item inside
- Close door
- Turn on the power
- Place hands inside the gloves
- Grab the blast hose, or suction gun
- Step on the foot pedal
- Blast surface
- Blow-off item with blow-off nozzle (inside cabinet)
- Inspect
- Repeat blast if needed
- Open door
- Remove item

Hand Sand

The hand sand method is widely used since sandblasting has been banned by some brands. Hand sand creates a very similar design as sand blasting. In hand sand, the worker uses emery paper to scarp away the top layer of indigo from the fabric. It can be used in particular placements and designs. The location chosen for the scraping can be front thigh or back seat. The most important factors in the hand sand are the right selection number of sand paper (in apparel industry, 220, 320 and 400 number papers are most popular and widely used), intensity and the pressure of scraping.

The process can be done by placing the apparel by using the rubber balloon or plain wooden boards. The hand sand must be started from intense part and gradually feathering out on less intense part. However, manual sandblasting is preferred by factories as it is cheaper because it does not require investment in advanced and expensive industrial equipment.

Sand Blasting & Hand Sand

Sand Blasting	Hand Sand
This process is done by a high-pressure air gun to shoot abrasive materials at fabric.	This process is done manually with the help of abrasive paper.
More consistent and uniform effect as it is done by a machine.	More irregular and varied effect as it is done manually.
This is a faster process.	This is a lengthy process.
Production is high.	Production is low.
More cost-effective than hand sand.	More time-consuming and expensive.
Can create a more uniform look for all fashion styles.	Can create unique and personalized designs on each garment.
Environment is polluted.	Environment is not polluted.
Health hazardous.	Less health hazardous.
Easy to control.	Difficult to control.

Why Sandblasting is Banned

Sandblasting is considered one of the most hazardous jobs due to the high exposure rate of toxic silica in the air during this work. This silica can be life-threatening if proper safety precautions are not put in place to protect employees from dangerous exposure.

Sandblasting was banned in the garment industry because it poses a significant health risk to workers who are exposed to the fine dust generated during the process. The dust contains silica, a mineral that can cause a range of health problems, including lung cancer, silicosis, and other respiratory diseases. The ban aimed to protect the health and safety of workers in the garment industry and to promote the use of safer and more sustainable methods for achieving the desired denim effects, such as laser fading and enzyme washing.

Silicosis

Silicosis is a lung disease caused by inhaling crystalline silica dust over a prolonged period. Silica is a mineral found in many rocks, including quartz, sandstone, and granite, and is commonly used in various industries, such as construction, mining, and manufacturing. Doctors often refer to silicosis as sandblaster’s disease. Once contracted, a patient’s lungs begin to fill with a fibrous tissue. Even after the patient ceases his exposure to the silica, his lungs continue to fill with the fibrous tissue.

When workers inhale silica dust, it can accumulate in the lungs and cause inflammation, scarring, and damage to the lung tissue. Over time, this can lead to symptoms such as shortness of breath, coughing, chest pain, and fatigue. In severe cases, silicosis can cause respiratory failure and can be fatal.

Precaution of Sand Blasting

Sandblasting generates a large amount of fine dust that can pose a significant health risk to workers if inhaled. Therefore, it is essential that workers performing sandblasting work use appropriate personal protective equipment (PPE) to protect themselves from dust and other hazards associated with the process. The following are some examples of protective equipment that can be used during sandblasting-

- Respirators

Respirators are designed to filter the air that the worker breathes, protecting them from inhaling dust and other airborne particles.

- Protective Clothing

Workers should wear protective clothing, including a full-body suit and gloves, to protect their skin from contact with abrasive particles and other hazards.

- Eye and Face Protection

Workers should wear goggles or a full-face shield to protect their eyes and face from flying debris, dust, and other hazards.

- Hearing Protection

Sandblasting can generate high levels of noise, so workers should wear earplugs or earmuffs to protect their hearing.

- Footwear

Workers should wear steel-toed boots or shoes with slip-resistant soles to protect their feet from heavy objects and to provide good traction on slippery surfaces.

3D Wrinkle



Wrinkle is a fold, ridge or crease in the cloth or garments. 3D wrinkle is a technique used in denim washing to create a three-dimensional, wrinkled or creased appearance on the fabric by using resin. Resin makes crosslinks between cellulosic bonds; when the fabric deforms, folds or losses its normal orientation, resin chemicals prevent the transformation to normal position and that causes wrinkles or creases. The 3D effect and natural vintage look adds value to denim garments and increases demand in the market.

Recipe for Resin Solution

- Resin (DMDHEU - Dimethyl Dihydroxy Ethylene Urea): 20%
- Water: 80%
- Softener: 0.5 gm/L (if required)

Process

At first the denim pant in set into the dummy. Some part of upper thigh of the pant is folded with clip according to design. Then the resin solution is sprayed with nozzle at measured quantity. After that, the garment is loaded in the oven for curing at 120º-140ºC for 20-40 minutes. When curing is complete, the garment is unloaded and left at room temperature for 10-15 minutes to cold dry.

Principle of Wrinkle Formation and Resin Action

Cotton is a cellulosic fiber and its polymer is linked by many hydroxyl (-OH) groups. The structural units of cellulose contain crystalline region, amorphous region and intermediate region. In the crystalline region, the cellulose chains are closely packed and the mobility of the chains are low. However, in the amorphous and intermediate regions, the molecular chains are temporarily held together with weak hydrogen bonds and the bonding can be broken easily when distortion force is applied.

After the force is applied, the temporary bonds would reform into a new position and the chains fail to return to their original positions. As a result, wrinkle or crease is formed. The resin forms covalent bonds to replace the weak hydrogen bonds between the cellulose chains. Therefore, the stability of the bonding would be improved, allowing the chains to hold onto the adjacent molecular chains more effectively.

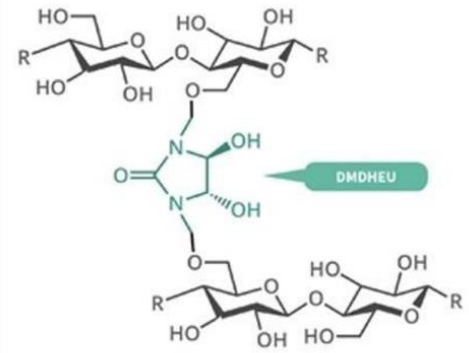


Figure: A cross-linking resin DMDHEU (Dimethylol dihydroxy ethylene urea) which forms covalent bond between cellulose chain that lock them in place result a permanent crinkle effect in the garment

Disadvantage of Resin

In this process DMDHEU works as the main cross-linking agent which links molecules of cotton fiber. Thus, creases appear on the garment. The hydrolysis of the N-hydroxymethyl groups from untreated cross-linking agent or from single end reacted with DMDHEU molecules may form formaldehyde. Formaldehyde is a toxic chemical that may cause irritation to eye, skin and throat; give an unpleasant odor; cause cancer to human body when taken for a long time, as well as respiratory problems and other health issues.

Moreover, the resin decreases the tensile and tear strength, abrasion resistance; may give unwanted harsh and stiff feel; turns the fabric yellowish after chlorine bleaching. Sometimes it is difficult to remove unfixed resin. If the resin is applied to elastic products, the elasticity of the fabric may be damaged, and for this reason the fabrics must be tested before applying resin.

Substitute of DMDHEU

Dimethyl dihydroxy ethylene urea (DMDHEU) is a type of formaldehyde-based water-soluble glyoxal resin. So, it may release formaldehyde, a known carcinogen, into the environment during production and use. The resin we are using can be altered with formaldehyde free substance that can perform similar crosslinking function. Such as- glyoxal with aluminum sulphate catalyst in high pressure can create 3D wrinkle on denim. Some of the substitutes for DMDHEU include:

- Crosslinking Agents

Crosslinking agents such as- glyoxal, dimethyl ethylene urea (DMEU) can be used as a replacement of DMDHEU. These are less harmful than DMDHEU and can create a similar effect on the fabric.

- Resin Blends

Resin blends that do not contain formaldehyde can be used as a substitute for DMDHEU.

- Organic Acids

Organic acids such as- citric acid, tartaric acid, and glycolic acid can be used as a replacement for DMDHEU. These organic acids work by breaking down the fibers of the fabric to create a natural, wrinkled look.

- Heat

Heat can be used to create wrinkles in denim by causing the fibers to shrink. This method is often used in conjunction with other finishing agents, such as- resins or softeners.

- Enzymes

Enzymes such as- cellulase can be used to create a similar distressed wrinkled effect on the fabric without the use of chemicals.

Whiskering



Whiskers (also known as 'moustaches', 'hige' or 'crease lines') are fade lines on the upper thighs, back of knees, and bottom of the denim pant legs. It is done on raw / rigid / unwashed denim.

The technique involves creating diagonal or horizontal creases or wrinkles on the fabric, mimicking the natural wear patterns that would occur over time with regular use. It is done with the help of sharp edge emery paper rolled on a fine wood stick or pasted on plastic material. Slim fitting jeans tend to have tight, straight whiskers, while looser fitting jeans usually have wide, more diagonally angled whiskers.

Process

At first the design (whiskering pattern) is selected. The selected pattern is drawn in garments using a pencil or chalk. Then garment is set in dummy/wooden board. Rubbing is done with emery paper or wire brush according to pattern. Emery paper is folded to produce thick and thin places according to pattern width. Then required wash is performed.

Tagging



Tagging or tacking is a process which is used to get heavy contrast effect on waistband, bottom hems, back pocket & front pocket corners etc. of denim garments. Tagging is done at unwashed (raw / rigid) condition of garments. After wash the effect is visible. Usually tag gun or plastic staple machines are used to attach plastic or nylon tag pins to garments. It can also be done manually by needle and thread.

Garment is folded on specified areas and the fold is locked by tag pins. Then garment is processed with specific wash and tag pin is removed after drying. The inner of the fold will be dark in shade due to less exposure to mechanical rubbing and chemicals. Variation is provided by using different lengths of tag pins varying from 5 mm to 15 mm.

Pin holes are sometime a problem to this process. This damage can be prevented by increasing the number of folds in one pin or by selecting the size of pin which is not loses to fold. It is recommended some time to pull off the tag pins before softening process. This help to prevent the pin holes but by doing so garment loses the folding effect after drying and only contrast remains. If folding effect is also required along with shade contrast, then pins must remain attached till garment is dried off.

Process

At first garment is marked according to design. Particular area of garment is folded as per design with tag pin. Then other washing process is performed keeping the tag pin attached. Then garment is washed with water and dried. Finally tag pins are opened.

Grinding



Grinding is the destruction of edges or the worn out effect on the edge of the garments. The process involves physically abrading the surface of fabric with a machine that has abrasive surfaces, such as sandpaper, stones or wire brushes, causing the fibers to break and resulting in a frayed and uneven appearance. Grinding machine with 2500 - 3000 rpm is used to destroy the edges.

It is often done on hem, pocket edge, waist belt, and pocket opening of the garments. This is a manual process and even with extreme care and control there is variation in results. Every garment will look unique and different form the others which is the beauty of vintage garment.

Destroy



Destroy is a technique that is used to create a heavily distressed and worn-out look on the fabric. The process involves physically breaking down the fabric, creating holes, frayed areas to give the fabric a rugged and aged appearance.

It can be achieved through various methods, such as sanding, scraping, cutting, or ripping the fabric, and is often combined with other techniques such as whiskering or grinding to create a more unique and individualized look.

After destroying process, the white yarn losses strength and to strengthen those yarns acetic silicon seal can be used which may prevent the breakout and tearing of those yarns. Or to reinforce the newly created holes, the garment is turned inside out and patch or cut piece of fabric is sewed around the hole.

There are three main methods for performing the destroy effect: automatic, semi-automatic, and manual.

Automatic: Upper surface dye is destroyed using CO_2 laser beam. The machine is programmed to create a specific pattern or style. It is faster than any other methods, and it can produce a consistent result. However, the consistent look is not preferable by buyer as it doesn't have any uniqueness.

Semi-Automatic: In this method, destroy effect is created using grinding pen and precisely cut the fabric using it. It allows for some degree of customization and create unique look on each garment.

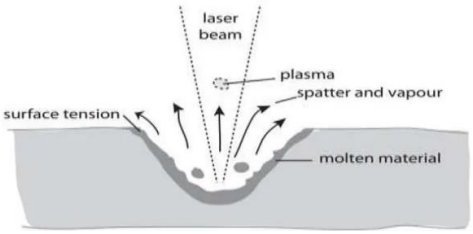
Manual: In this method, destroy effect is created using a sharp knife and fabric surface is cut slightly as slice. Warp yarns are cut thru knife and weft yarns are kept to show white thread. Also, holes can be made by cutting warp and weft yarns in torn jeans. This is the most time consuming method, but allows for most customization and control. For this manual process, every garment will look unique and different than others which is preferred by buyer.

Laser Fading



Laser fading is a popular dry process for denim these days. It has been used extensively as the replacement of some conventional dry processes like sand blasting, hand sanding, destroying, and grinding etc. which are potentially harmful and disadvantageous in some manner. By using a laser, it is possible to create a worn out look on denim, which could be an alternative method to conventional processes. Generally, a CO_2 laser is a suitable tool for the discoloration of indigo dye on denim fabric.

Process



In the laser-based fading process, a computer drives the laser beam to the material where marking or fading is required. The laser beam focuses on a small area of the fabric, causing it to heat up, decomposes the dye and the resulting vapors are vented away. The material fades only where the beam impacts on the fabric. Commercially two types of lasers are being used: solid based (wavelength of 1 µm) and gas based (wavelength of 10 µm). The desired degree of fading depends upon the wavelength, power density, pulse width, and duration of the laser beam.

Specifications of CO_2 Laser

- Average output power 150-250 watts
- Peak output power 230-400 watts
- Working frequency 50-60 Hz
- Linear polarization
- Wavelength 1 µm (solid based) -10 µm (gas based)
- Pulse width 1-100 nanoseconds

Limitations

- Cannot mimic unique wear patterns and color variations that develop on denim organically.
- Cannot replicate natural aging effects alone that occur over time with repeated wear and wash.
- Need extra manual wash to create unique look.

Advantages

- Any design at any place can be created.
- Fading on seams, or metal buttons is possible.
- Faster process compared to the conventional dry processes.
- Almost 100% design accuracy.
- Suitable for wide range of material.
- Low hairiness compared to other mechanical fading.
- Environment friendly.
- Zero water waste.
- Requires no chemicals.
- Less manpower required.
- Comparatively less strength loss than other mechanical fading. For example, in a study it was found that, for hands sand brush tensile strength in warp ways decreased by 56.68%, for laser faded tensile strength decreased by 1.31% and for PP spraying reduced by 2.57% in comparison with the raw denim.

Disadvantages

- Initial investment is very high, which is a problem for small and medium size industries.
- Skilled operator is needed, which might be a disadvantage in some particular circumstances.
- Producing natural effect is difficult and often requires some manual touch after the fading to give the product natural look.
- The laser beam itself and the produced fumes are health hazards.
- Maintenance and servicing of the laser system may be troublesome in some circumstances.

Why Laser Fading is a Sustainable Wash

By reducing chemical usage, minimizing water consumption, optimizing energy usage, and improving occupational safety, laser fading contributes to a more sustainable approach to denim washing.

- Reduced Chemical Usage

Laser fading significantly reduces the need for chemical treatments that are typically used in traditional fading processes. It minimizes the use of harsh bleaching agents, enzymes, and other chemicals that can be harmful to the environment and human health.

- Lower Water Consumption

Traditional fading methods often require multiple rinses and large volumes of water to remove chemicals and achieve the desired faded effect. Laser fading reduces or eliminates the need for excessive water usage in the fading process.

- Energy Efficiency

Laser machines are generally energy-efficient compared to other industrial washing processes, contributing to reduced energy consumption and lower carbon emissions.

- Precise and Controlled Process

Laser fading offers precise control over the fading process, allowing for targeted and localized fading effects. This precision helps to minimize wastage and optimize the use of resources.

- Waste Reduction

Laser fading minimizes the generation of waste materials as it reduces or eliminates the need for abrasive stones, pumice stones, or sanding materials that generate significant waste during the fading process.

- Improved Occupational Safety

Laser fading reduces the exposure of workers to hazardous chemicals used in traditional fading methods. It eliminates the need for manual labor-intensive processes, such as hand sanding or stone washing.

Apparel Dyeing



Apparel dyeing is the process where garments are made from grey fabric and then dyed in the required color and shade. It was first used for fully fashioned garments which are dyed after the manufacturing process is completed.

Advantages

- Just in time production facility.
- Any color and shade can be produced at any time.
- Can create new fashion trends by combining washing and dyeing.
- Less possibility of shade variation within the garments.
- Old garments can be redyed, hence becomes like new garments.
- Manufacturers can stock 'white styles' that can be dyed or printed later as per order.
- Desizing, scouring, bleaching, dyeing, and finishing con be done in the same machine.
- Reduce lead time, increase production efficiency for both fabric and garment producers.
- Comparatively lower cost of production.
- Comparatively lower capital investment.
- Requires a lower liquor ratio, hence lower garments dyeing cost.
- During fabric cutting approximately 10-15% of the fabric is cut out as wastage, if this fabric is cut out as grey fabric rather than colored fabric, then the wastage cost will be saved.

Disadvantages

- High cost of processing.
- More material handling.
- Special care for selection of interlining.
- Higher seconds rate in production.
- Labor intensive process.
- Requires full checking of all pieces.

Precautions

- Dyeing should be done after pre-treatment to reduce problems.
- Garments design should be made from engineering point of view.
- Garments which are produced from woven and knitted fabrics combinedly, the knit fabric should be pre-shrunken to avoid seam puckering.
- If the garment fabric is of mixed fibers, dye should be taken carefully, having equal dye pick-up to avoid irregular dyeing.
- Garments made from compactly woven cotton fabric should be dyed by hot brand reactive dye.
- Sewing thread should be of same fiber like the garment fabric.
- Seams should not be too tight or too loose especially in the areas containing elastic like waistband, cuff etc. which may create irregular dyeing.
- Nylon and rubber type button may be destroyed in the dyeing process. Polyester button is better in the regard.
- Elastic should be of polyester type or dye should be copper free. Rubber or polyurethane type elastic may be damaged by the action of temperature and chemical.
- If any metal component is used, should be made from nickel or nickel alloy because poor quality metal may be damaged by the action of salt or alkali.

Special Considerations

- Fabric should be prepared according to shade. For light shade, pre-treatment is not necessary. For dark shade, pre-treatment is necessary to clean the fabric and increase dye absorbency.
- If possible, garments should be made from the grey fabrics of the same lot and same source.
- Pre dyed or dyeable buttons should be used.
- Polyamide plastic zippers should be used which are suitable for dyeing.
- For 100% polyester thread, it should be pre dyed matching with shade or core spun cotton thread can be used.
- Interlinings recommended for garments dyeing should be used, otherwise the performance of interlining may be destroyed.

Fully Fashioned Garments

Fully fashioned garment means that the fabric is knitted into final shape by a knitting machine and then sewn together. Typically made on circular knitting machines, which allow the fabric to be knitted in a continuous loop. This process results in garments that are seamless and have a more comfortable fit. Examples- sweaters, socks, lingerie, swimwear, sportswear, etc.

A semi-fully fashioned garment is a garment that is partially knitted in its final shape and partially cut and sewn. Typically made on flatbed knitting machines, which allow the fabric to be knitted in a series of panels. These panels are then cut and sewn together to create the garment. This process results in garments that are more affordable than fully fashioned garments, but they are not as seamless or as comfortable.

Garments Dyeing & Fabric Dyeing

Garments dyeing can be more cost-effective in terms of dye usage and resource consumption. This method involves applying the dye directly to the finished garments, resulting in precise color control and minimal dye waste.

On the other hand, fabric dyeing requires dyeing larger quantities of fabric, which lead to higher dye consumption. It involves pre-treatment processes like desizing, scouring, and bleaching to prepare the fabric for dyeing. During fabric cutting, about 10-15% of the fabric is wasted. By cutting this wastage as grey fabric instead of colored fabric, additional labor, machinery, and process costs can be saved.

Here are some differences between garments dyeing and fabric dyeing-

Garments Dyeing	Fabric Dyeing
Readymade garments are subjected to dyeing.	Grey fabrics are subjected to dyeing.
Cost is lower.	Cost is comparatively higher.
Less possibility of shade variation.	Higher possibility of shade variation.
Require lower liquor ratio.	Require higher liquor ratio.
Possibility of old garments re-dyeing.	Less possibility of fabric re-dyeing.
Desizing, scouring, bleaching, dyeing, finishing can be done in same machine.	These processes are not possible in the same machine.
Investment cost is lower.	Investment cost is higher.
Dyeing process require less time.	Dyeing process require more time.
Lower waste generation.	Higher waste generation.
Environmental impact is less.	Environmental impact is high.
Color matching and quality control can be more challenging due to individual garment dyeing.	Color matching and quality control is comparatively easier.

Responsibility of Technical Officer in Garments Washing and Dyeing

- Solve complaints.
- Root cause analysis and solving.
- Analyze floor recipes, check and find out high costing recipes, technical fault factors.
- Trial new batch/lot for process improvement, reduce costing.
- Developing and implementing new washing and dyeing techniques.
- Develop and implement standard operating procedures (SOPs).
- Overseeing the safety and environmental compliance of the washing and dyeing process.

Responsibility of IE Executive in Garments Washing and Dyeing

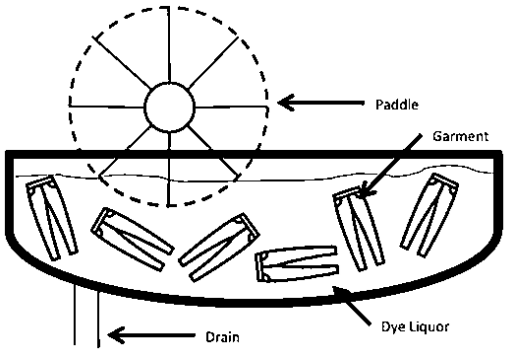
- Process analysis and improvement.
- Production planning and scheduling.
- Workstation design and layout.
- Standardization and documentation.
- Time and cost estimation.
- Quality control and inspection.
- Developing and implementing process improvement initiatives.

Dyeing Machines

All apparel dyeing machines can be classified into two groups based on their construction- pedal dyeing machine and rotary dyeing machine.

Pedal Dyeing Machine

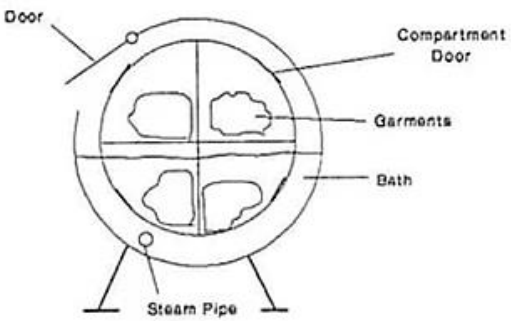
Pedal type garments dyeing machine is the first and oldest type of garments dyeing machine still used in the European countries. Here, garments are gently moved using paddles similar to a paddle wheel on a boat. This is a slow process, but there is extremely little abrasion on the goods.



These machines are used for dyeing woolen knit wear type garments. Dyeing cost is higher in these machines due to manual operation, higher labor load, manual loading and unloading of garments, manual chemical feeding, very high liquor ratio etc. It is not used in Bangladesh.

Rotary Dyeing Machine

Rotary garment dyeing machine is the modified version of pedal type garment dyeing machine. These machines are of two types: normal atmospheric pressure type (less than 100ºC) and high temperature type (more than 100ºC).



These machines move the garments and liquor in the machine with a perforated rotating drum. The outer drum or roller is positioned fixed to the floor. The inner drum is perforated. The inner drum can rotate both clockwise and counterclockwise.

Dyes

Direct Dyes: It is not used in factories due to low color fastness to wash. Mostly used in boutique. A care label of "hand wash separately" should be attached if used in garments.

Reactive Dyes: Best suitable for cotton garment dyeing due to its good level dyeing properties and high migration, diffusion properties of both cold brand and hot brand. It is the only dye that reacts with the textile fiber and makes covalent bond with fiber. The dye becomes a part of the fabric and gives an outstanding color fastness to wash and light.

Vat Dyes: It has limited use in garments dyeing.

Sulphur Dyes: Sulfur dyes are usually used for black, dark blue (navy), and brown color.

Process Flow

Desizing	Garment Loading → Machine Run → Add Water → Chemical Dosing → Normal Run (2 minutes machine run for mixing) → Temperature Raise → Run Time → Drain → Normal Wash → Drain
↓	
Dyeing	Add Water → Glubar Salt (exhaustion) → Normal Run → Color Dosing → Normal Run → Soda Ash (alkaline medium) → Normal Run → Temperature Raise → Run Time → Drain → Normal Wash → Drain
↓	
Soaping	Add Water → Chemical Dosing → Temperature Raise → Run Time → Drain → Add Water → Normal Wash → Drain
↓	
Fixing	Add Water → Chemical Dosing → Temperature Raise → Run Time → Drain → Add Water → Normal Wash → Drain
↓	
Softening	Add Water → Chemical Dosing → Run Time → Drain
↓	
Hydro Extracting	
↓	
Drying	
↓	
Quality Checking	

Dyeing Recipe

Washing Machine RPM: 10, Dryer RPM: 15, M:L = 1:15, Product: 100% Cotton Sweatshirt

Process	Description	Process Time	Chemical %	Temperature
Desizing	Garments Loading			
	Machine Start			
	Water			
	Chemical Dosing		1 gm/L	
	Normal Run	5 min		
	Temperature	5-7 min		60ºC
	Run Time	20 min		
	Drain			
	Normal Wash - 1	1-2 min		
	Drain			

Dyeing	Water			
	Glubar Salt Dosing	2-3 min	50 gm/L	
	Normal Run	5 min		
	Color Dosing	5-7 min	1.50%	
			2%	
	Normal Run	5 min		
	Soda Ash		20 gm/L	
	Normal Run	5 min		
	Temperature	5-7 min		80ºC
	Run Time	30 min		
	Drain			
	Normal Wash - 2	1-2 min		
	Drain			
Soaping	Water			
	Chemical Dosing		1 gm/L	
	Temperature	2-3 min		60ºC
	Run Time	10 min		
	Drain			
	Water			
	Normal Wash - 1	1-3 min		
	Drain			
Fixing	Water			
	Chemical Dosing	2-3 min	4.5 gm/L	
	Temperature	3-5 min		50ºC
	Run Time	10-15 min		
	Drain			
	Water			
	Normal Wash - 1	1-2 min		
	Drain			
Softening	Water			
	Chemical Dosing	2-3 min	1 gm/L	
	Run Time	5 min		
	Drain			
Hydro Extractor	Hydro	5 min		
	Garments Unloading			
Drying	Dryer	30-40 min		
Quality Check				
Delivery				

Dyeing Parameters

Chemical Concentration

Chemical concentration refers to the amount of dye or other chemical additives used in dyeing process. It can affect color depth, shade, and colorfastness. Ideal concentration will vary depending on the type of dye being used and the desired results.

Temperature

Temperature of dye bath affect dye absorption, migration, and fixation. Higher temperatures will result in a faster dyeing process, but they can also damage the garment. The ideal temperature will vary depending on the type of dye being used and the desired results.

Time

Longer dyeing times will result in a deeper color, but they can also damage the garment. The ideal time will vary depending on the type of fabric, type of dye being used and the desired results.

M:L Ratio

M:L ratio influences dyeing efficiency, level of color, and resource consumption. Lower M:L ratio generally results in higher color intensity but may require more dye and energy. Higher M:L ratio ensures even distribution of dye and other chemicals leading to uniform dyeing.

Machine RPM

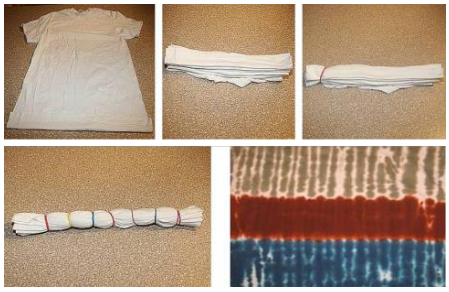
It influences the degree of agitation and mechanical action applied on the garment in dye bath. Higher RPM can enhance dye penetration but may also cause abrasion related damage. Lower RPM gently move the garments inside machine and create better quality dyeing.

Fashion Dyeing

Tie Dye

Tie dye is a dyeing technique that involves creating unique and colorful patterns by tying or binding sections of fabric before applying dyes. The process typically involves folding, twisting, pleating, or crumpling the fabric and securing it with rubber bands, or strings to create different patterns and designs. The tied fabric is then immersed or sprayed with dyes of various colors. It can be done on a variety of fabrics, including cotton, silk, rayon, and other natural or synthetic fibers.

- Stripes



The fabric is rolled very loosely, forming a long tube. The stripes will be at right angles to the tube. The fabric is tied at one-inch intervals, spaced according to the desired width of the stripes. Impermeable bands or string are looped around the tube a few times and knotted. The ties should be made very tight. Now, the fabric can be immersed in tie-dye water.

- Random Circles



Random circles effect is made by tying knots with string, thread or elastic bands in different places. The more fabric is tied, the larger the circles. The less fabric that is tied, the smaller the circles.

- Spirals



Spiral design is created by arranging pleats of fabric to spin around a central point, gathered into a round bundle. Different slices of the circular bundle are usually dyed different colors. The fabric is first laid out on a flat surface. The thumb and fingers of one hand are placed together on the cloth where the center of the design is desired. The fabric is then twisted so that a spiral is created. After each twist, the fabric is flattened to maintain its flatness and prevent folds from rising or lumping up. Finally, the loose ends are brought into the circle, resembling a pancake, and secured with bands or string, intersecting like spokes on a wheel.

- Crumpled



The t-shirt is laid flat on a work surface and gathered up into random wrinkles by spreading the fingers out and crumpling it into a disc shape. The goal is to expose as much of the t-shirt to the dye as possible, so it is important to make sure that there are many small folds and wrinkles. Once the shirt has been crumpled, 3 or 4 impermeable bands are wrapped around it. Dye is then applied randomly to the t-shirt using squirt bottles.

Dip Dye



Dip dye is a dyeing technique that involves submerging part of a garment in dye, creating a gradient effect. In this process, the garment is partially submerged into a dye bath, resulting in a gradual color transition from the immersed portion to the unimmersed portion.

Here's a step-by-step explanation of the dip dyeing process:

- The dress is laundered to eliminate any finishes that might hinder the absorption of dyes.
- A plastic cover or tablecloth is used to cover the work surface.
- Dye bath is prepared in a disposable container, mixing ½ cup liquid dye or 1 box powder dye with 1 cup salt and 4 cups very hot water, and stirred thoroughly.
- Dye solution is poured into a 3-gallon container, adding 2 gallons very hot tap water and a squirt of liquid detergent, and stirred thoroughly.
- Dress is wetted and excess water is squeezed out. Two-thirds of the dress is wrapped around a dowel rod. Bottom one-third of the dress is immersed in the dye bath for 10 minutes, with gentle movements back-forth and up-down.
- Next, middle section of the dress is dipped into the dye bath for 5 minutes. It is gently lifted and lowered to create varying shades, ranging from lighter to darker.
- Then top section of the dress is unrolled, and quickly dipped into the dye bath just long enough to obtain a very light shade.
- Dress is removed from dye bath and excess dye is wring out. Dress is then rinsed under cool running water, holding the lightest color at the top for allowing dye to flow through to the darkest area.
- Subsequently, the dress is rinsed in warm water until the water runs clear, washed with detergent in warm water, thoroughly rinsed, and excess water is wring out. Finally, either machine dried or hung to dry.

Over Dye



Over dyeing is a dyeing technique in which a fabric or garment is dyed with a new color over an existing color or dye. It is used to change or modify the original color of the fabric, creating a different and often more vibrant hue.

Over dyeing allows for a range of creative possibilities, including enhancing or intensifying the original color, creating new color combinations, or completely transforming the appearance of the fabric or garment. The success of over dyeing depends on various factors, such as the fabric type, dye compatibility, dyeing technique, and skill in color mixing and application.

Tinting



Tinting refers to the process of adding a small amount of color to a base color or material to create a lighter or pastel shade. After denim pant is being faded, the pants are dyed with very light color (0.001% or 0.002% yellow or pink). This dyeing process is called tinting. The process changes the hue/cast/tone of indigo, gives a used/vintage and muddy look.

This is not true over dyeing as it merely gives the impression of a change in the overall color of the fabric. As soon as quantity of tint color increases and it cover up indigo, reaches the level of dyeing. It is applied only on garments, not in fabric or yarn. Traditionally tinting is done with direct dye or reactive dye.

Apparel Printing

Apparel printing refers to the process of applying colors, designs, patterns, or graphics onto localized area of clothing items or garments. Apparel printing encompasses a wide range of methods, including screen printing, direct-to-garment (DTG) printing, heat transfer printing, sublimation printing etc. Each technique has its own set of advantages. The choice of garment printing method depends on factors such as- complexity of the design, type of garment, desired quantity, and budget.

Dyeing & Printing

Dyeing	Printing
Thorough coloration process of textile materials.	Localized application of dyes and pigments.
Color is applied in form of solution.	Color is applied in form of thick paste.
Color penetrates through the fabric.	Color is applied only on the fabric surface.
Whole fabric is uniformly covered with one color.	One or more colors are applied only to defined areas to obtain the desired pattern.
No use of thickener.	Thickener is used.
Steaming is not required on dyed fabric.	Steaming is used in the printed fabric for fixation.
More time is required in the batch applications.	Less time is required in continuous process.
Huge amount of water is required.	Less amount of water is required.

Rubber Printing

Rubber printing is a type of printing that uses a rubber-based ink to create raised, textured designs on garments. The concept behind it is to create a raised, 3D-like effect on the fabric by applying a thick layer of rubber-based plastisol ink onto the garment. When the ink is heated to a high temperature, it cures and solidifies, forming a raised, rubbery texture. This not only adds a tactile element to the design but also provides excellent opacity and vibrant colors.

Features



- Raised texture.
- Durable and long-lasting.
- Vibrant colors.
- Suitable for various fabrics.

Procedure

- Design or artwork to be printed is prepared in a digital format.
- A screen is prepared for each color in the design.
- Plastisol ink is applied to the garment using a screen printing press.
- Garment is dried or heat pressed to cure ink, causing it to solidify and form rubbery texture.
- Then garments are cooled down, and final prints are inspected for quality and consistency.

High Density Printing

High density printing is a printing process that involves stacking layer upon layer of ink to raise the print above the fabric by up to 3 mm (1/8 inch).

The concept behind it is to use a thick and raised ink to create a three-dimensional effect on the fabric. It involves using a special ink that expands when heated, resulting in a thick and raised texture on the fabric that adds depth and dimension to the artwork.

Features



- Raised texture.
- Durable and long-lasting.
- Bold and vibrant colors.
- High level of detail.
- High contrast between the design and the fabric.

Procedure

- Design or artwork to be printed is prepared in a digital format.
- A screen is prepared for each color in the design, including one for the high-density ink.
- High-density ink is applied to the garment using a screen printing press.
- Additional layers of ink are added to create the desired thickness.
- Garment is dried or heat pressed to cure ink, causing it to expand and form the raised texture.
- Then garments are cooled down, and final prints are inspected for quality and consistency.

Puff Printing

Puff printing is a type of printing that uses a special ink that puffs up when heated, creating a raised, 3D effect. Puff printing works by using a special ink that contains a heat-activated foaming agent. When the ink is heated, the foaming agent expands, causing the ink to puff upward. This creates a raised, 3D effect that is very visually appealing.

Features



- Raised Texture.
- Soft Feel.
- Durable and Long-lasting.
- High contrast between the design and the fabric.

Procedure

- Design or artwork to be printed is prepared in a digital format.
- A screen is prepared for each color in the design, including one for the puff ink.
- Puff ink is applied to the garment using a screen printing press.
- Garment is dried or heat pressed to cure ink, causing it to expand and puff up.
- Then garments are cooled down, and final prints are inspected for quality and consistency.

Transfer Printing

Transfer printing is a printing process in which an image or design is printed onto a transfer paper, which is then applied to a fabric or other material using heat and pressure. When heat and pressure are applied, the ink is released from the paper and bonds with the fabric, creating a permanent and vibrant print.



Features

- Highly detailed and intricate designs with fine lines, gradients, and complex patterns.
- Enables printing with multiple colors and shades, allowing for photo-realistic prints.
- Versatile for use on different types of garments.
- Resulting print can have a soft feel, making it comfortable to wear.
- Properly executed transfer prints are durable and able to withstand repeated washings.

Procedure

- Design or artwork to be printed is prepared in a digital format.
- Design is printed onto a transfer paper in reverse.
- Transfer paper is then placed on the fabric with the design facing down.
- Heat press is then set to the correct temperature and pressure.
- Heat press is then applied to the fabric for the correct amount of time.
- Then garments are cooled down, and transfer paper or film is peeled away.

Sublimation Printing

Sublimation printing is method of transfer printing. It is a process of transferring sublimable dyes from a transfer paper onto synthetic fabrics, such as polyester, using heat and pressure.

The concept behind sublimation printing is based on the sublimation process, where a solid dye is transformed into a gas without passing through a liquid state. The sublimation ink is printed on a special transfer paper in reverse, and when heat and pressure are applied, the ink turns into a gas and permeates the polyester fabric, bonding with the fibers and becoming a permanent part of the garment.



Features

- Most effective on synthetic fabrics.
- Vibrant full color prints.
- Long lasting.
- Breathable and comfortable to wear.
- Soft feel.

Foil Printing

Foil printing is a decorative method used to add metallic or reflective designs to garments. Foil printing works by using a heat press to apply heat and pressure to a piece of metallic foil that has the desired design on it. The heat and pressure cause the foil to melt and transfer to the fabric. The paper is then peeled away, leaving the shiny metallic design on the fabric. Foil printing is commonly used for adding a touch of luxury or creating a premium look on clothing items.



Features

- Adds a touch of luxury or sophistication to garments.
- Available in a wide range of colors and finishes.
- Provides a metallic or reflective appearance to the design.
- Offers a high contrast between the foil and the fabric.
- Durable and long lasting if applied correctly.

Procedure

- Design or artwork to be printed is prepared in a digital format.
- Desired color and type of foil is chosen based on the design and garment color.
- Foil adhesive is screen printed or applied to the areas where foil will be transferred.
- Foil is placed on top of garment, with colored side facing up and metallic side facing down.
- Heat and pressure are applied to activates the adhesive and transfer foil onto fabric.
- Then garments are cooled down, and foil is peeled away.

Flock Printing

Flock printing is a decorative method involves depositing small, finely cut natural or synthetic fibers (called flocks) onto the surface of the fabric with the help of adhesive.

The concept behind flock printing is to use an adhesive to create a pattern or design on the fabric. The flock, which consists of short fibers made from materials such as nylon or rayon, is then applied to the adhesive-covered areas. The flock adheres to the fabric, creating a soft and plush surface, giving the design a distinct, velvety texture.



Features

- Creates a furry or textured effect on garments.
- Variety of colors and finishes.
- Very soft to the touch.
- Durable process that can withstand repeated washing and wear.

Procedure

- Design or artwork to be printed is prepared in a digital format.
- Adhesive is screen printed or applied to the areas where flock will be transferred.
- Flock is gently pressed into the adhesive, ensuring that it adheres to the fabric effectively.
- Excess flock is removed, leaving the desired design with a velvety texture.
- Printed garment is then cured, either through heat or air-drying, to ensure that the adhesive sets properly.

Digital Printing

Digital printing is a process that involves printing designs directly onto the fabric using a digital printer. It is also called “direct-to-garment (DTG)” printing because the design is printed directly onto the fabric without any intermediate steps.



The concept behind digital printing is to utilize advanced inkjet technology to print designs directly onto the fabric without the need for separate screens or stencils. The printer sprays water-based inks onto the garment, and the ink is absorbed by the fabric, creating a permanent and high-quality print. Print is sharper and has a higher resolution or DPI than traditional printing methods. Digital printing is ideal for small orders and custom designs, but it may not be as cost-effective as other printing methods for larger orders.

Features

- Application of intricate and full-color designs, including gradients and photo-realistic images.
- Accommodate various garment types, including cotton, polyester, blends etc.
- Fast turnaround time, suitable for small orders, customizations, or on-demand printing.
- Water-based inks create a soft and smooth finish on the fabric.
- Water-based inks are more environmentally friendly compared to traditional printing.
- Durable and long lasting.

Procedure

- Design or artwork to be printed is prepared in a digital format.
- Garment is prepared by ensuring it is clean and flat.
- Garment is placed onto a flatbed or platen of the digital printer.
- Printer sprays water-based inks by print head directly onto the fabric according to the design.
- Garment is heat-cured using a heat press or conveyor dryer and ink is set.

Synopsis of DTG printer

Direct to garment printing in the United States began in 1996.

First commercially available DTG printer named “Revolution”, developed by Matthew Rhome of Bradenton, Florida.

The Revolution printer was offered for sale until 1998 when Rhome left the company to start development of the first Brother DTG printer, which came to market in 2005.

In 2005, at the ISS Atlantic City show, Brother International introduced the GT-541 Garment Printer to the market. This printer had print heads, ink, and electronics developed specifically for DTG printing.

At the Chicago PRINT 2013 show Epson introduced the F2000 printer. The release of this printer was notable as it addressed many of the issues prevalent in DTG printing at the time. One of the most important features of the Epson F2000 was its ink set as it had a two year shelf life and did not have the settling or clogging issues of previously introduced DTG inks.

Epson SureColor F2000 DTG Printer Working Process

- Design or artwork to be printed is prepared in digital format (by coreldraw/ adobe illustrator).
- Garment is laid in the pre-treator plate (for few seconds).
- Garment is dried in the heat press. Silicon paper is used to protect the heating element. Temperature 330ºC, time 20-30 second, high pressure to completely dry the pretreated area.
- The garment is placed in the print area by the help of the hoop.
- Machine will print a white under base, then the color. Time required is 2-3 minutes but depends on garment color, size and design.
- Then garment is cured in the heat press machine. Temperature 330ºC, time 90 second, light pressure is employed.

DTG & Screen Printing

DTG	Screen Printing
Unrivaled, photorealistic high quality prints.	Vibrant high quality prints.
Not cost-effective for extremely large batches.	Cost-effective for large batches.
Extensive color options.	Ideal for simple designs with limited colors.
Able to withstand over 50 washes.	The best choice for garment longevity.
Excellent for extremely detailed and intricate designs.	Difficult to print finer details; perfect for less detailed designs.
Typically uses environmentally friendly, water-based inks.	Screen printing can be dirty and messy, requires a lot of water.